

Chapter Two

Labor Supply

2.1. Measurement of the Labor force Participation

- **Labor force (LF)** = Unemployed + Employed
- **Unemployed:**
 - ✓ Aged 15/16 or older
 - ✓ temporary layoff from a job
 - ✓ Have no job, but actively looking for work
 - ✓ Applied/attempted to apply for a job in the four-week period prior to the data collection
- **Employed:**
 - ✓ Aged 15/16 or older
 - ✓ Worked at least an hour paid job per week or 15 hours or more (per week) on a nonpaid job (such as the family business).
- **Not in the labor force**
 - Aged 15/16 or older
 - Permanent layoff
 - Do not applied for or searched job in the last 4 weeks prior to the data collection
 - Gave up looking for a job
- **The labor force participation rate** = LF/P ; P = population size

2.1. Measurement of the Labor force Participation

- The **employment rate** (also called the “employment–population ratio”) gives the fraction of the population that is employed:
 - ✓ Employment rate = E/P
- The **unemployment rate** gives the fraction of labor force participants who are unemployed:
 - ✓ Unemployment rate = U/LF
- Which one of the above to use as a measure of the extent of the healthiness of the overall economy?
- Unemployment rate: - excludes the **hidden unemployed** - during severe recession, laid-offs become discouraged to look for job and considered as ‘out of labor force’, in that the unemployment rate underestimates the depths of the recession and economic hardships.
- Employment rate: it may decrease as school enrolment increases and as fertility increases → employment rate may not show the true economic performance.
- The most widely used measure of the labor market and the performance of the economy is the unemployment rate.
- Unemployment rate serves as a measure of the extent of the healthiness of the overall economy.

2.1. Measurement of the Labor force Participation

- It is however, important to note that there are sizeable flows of workers between groups. We can identify four major flows of people between groups:
 1. From the employed to unemployed groups: - some workers may be voluntarily quitting or laid off
 2. From the unemployed to the employed groups: - some unemployed workers may be newly hired or recalled from temporary layoff
 3. From the labor force to out of the labor force: - some employed or unemployed workers may retire or drop out of work
 4. From out of the labor force to the labor force (new graduates, for example) and some of the drop outs may reenter the labor force

Overtime labor force participation: USA – men

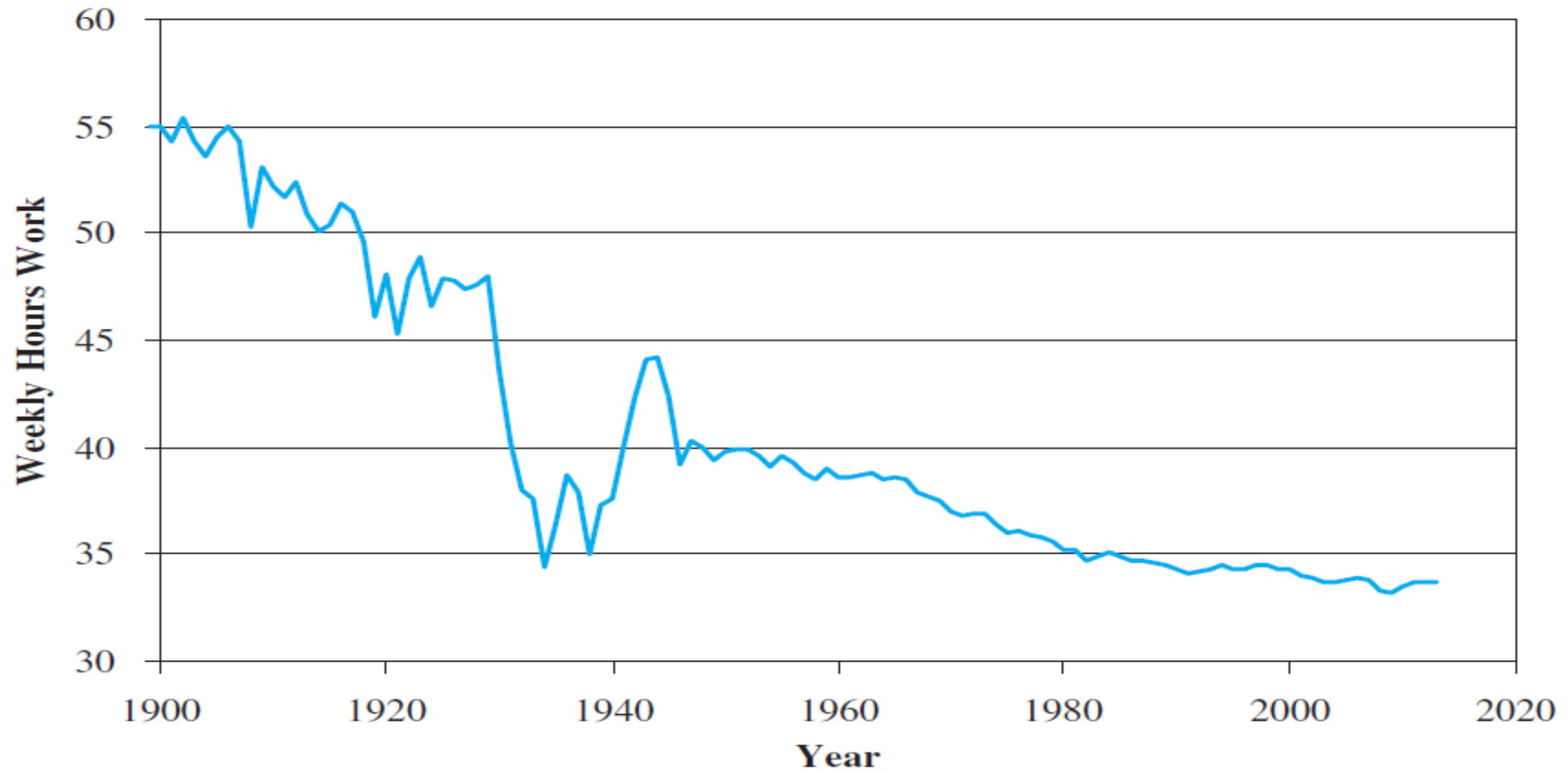
(Borjas, 2016)

Year	All Men	Men Aged 25–44	Men Aged 45–64	Men Aged over 65
1900	80.0	94.7	90.3	63.1
1920	78.2	95.6	90.7	55.6
1930	76.2	95.8	91.0	54.0
1940	79.0	94.9	88.7	41.8
1950	86.8	97.1	92.0	45.8
1960	84.0	97.7	92.0	33.1
1970	80.6	96.8	89.3	26.8
1980	77.4	93.0	80.8	19.0
1990	76.4	93.3	79.8	16.3
2000	74.8	93.1	78.3	17.5
2010	71.2	90.6	78.4	22.1

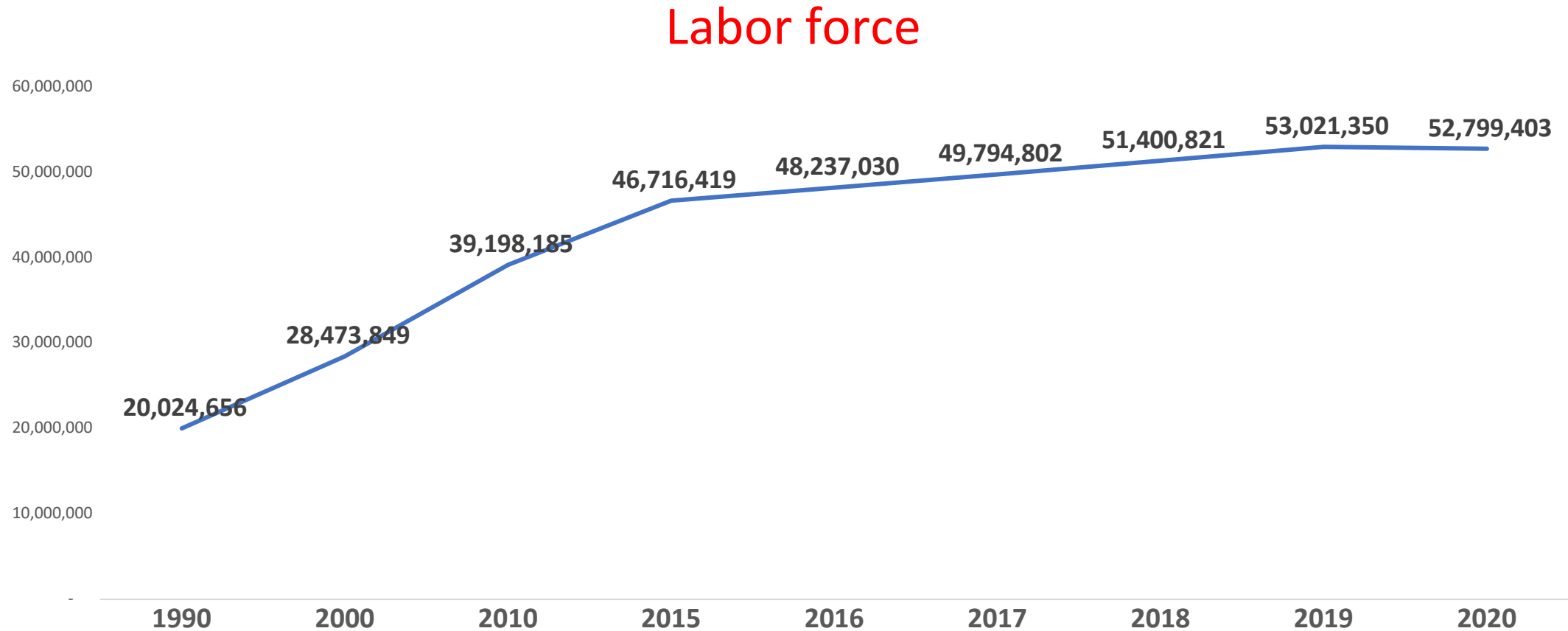
Overtime labor force participation: USA – women (Borjas, 2016)

Year	All Women	Single Women	Married Women	Widowed, Divorced, or Separated
1900	20.6	43.5	5.6	32.5
1910	25.4	51.1	10.7	34.1
1930	24.8	50.5	11.7	34.4
1940	25.8	45.5	15.6	30.2
1950	29.0	46.3	23.0	32.7
1960	34.5	42.9	31.7	36.1
1970	41.6	50.9	40.2	36.8
1980	51.5	64.4	49.9	43.6
1990	57.5	66.7	58.4	47.2
2000	59.9	68.9	61.1	49.0
2010	58.6	63.3	61.0	48.8

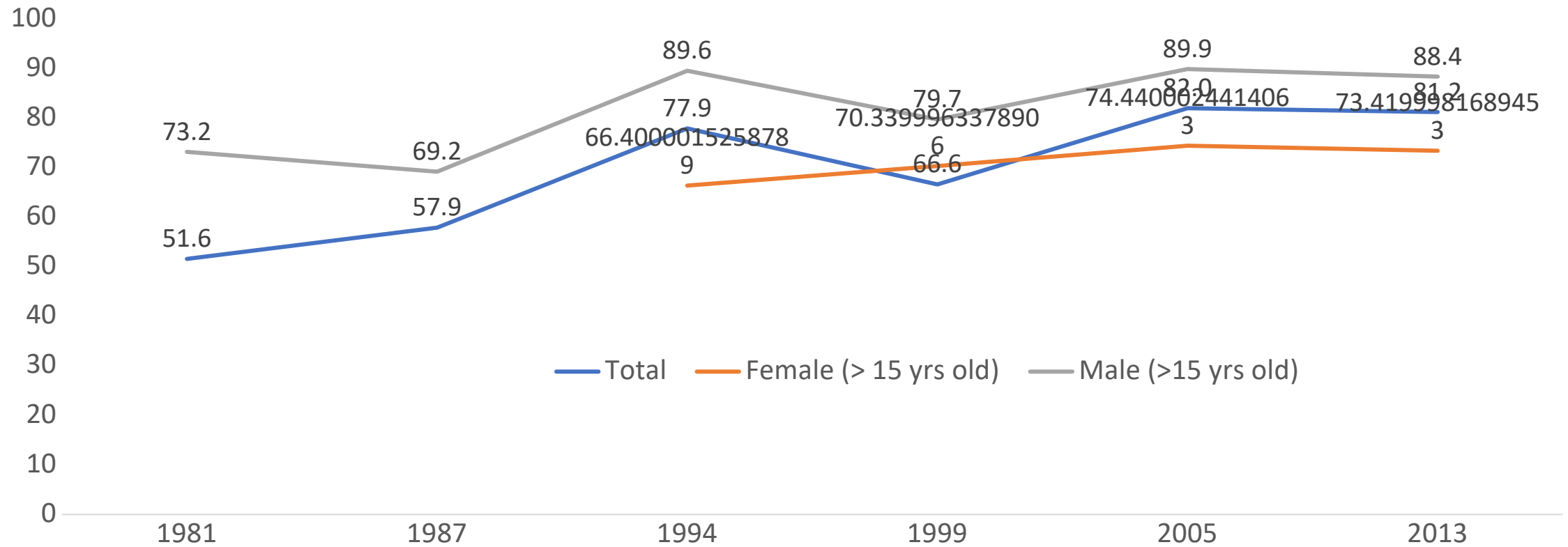
Overtime Working hours: USA (Borjas, 2016)



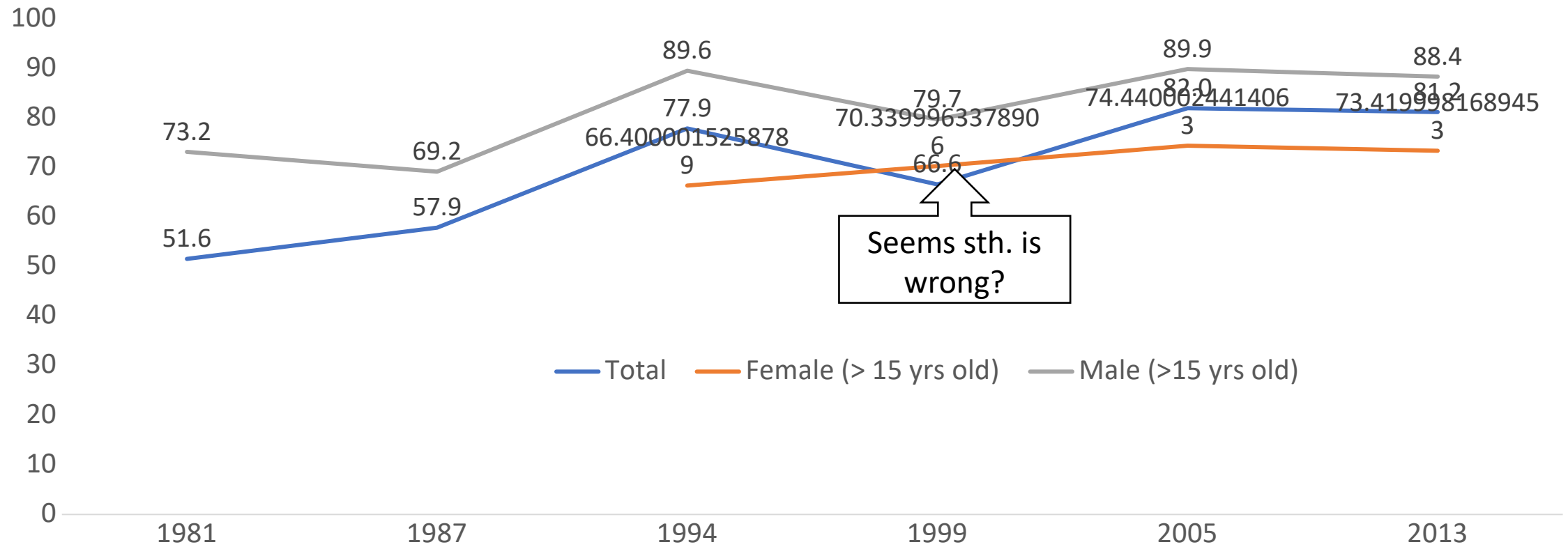
Overtime labor force : Ethiopia (compiled from the WB data)



Overtime labor force participation rate : Ethiopia (compiled from the WB data)



Overtime labor force participation rate : Ethiopia (compiled from the WB data)



2.2. The Work-Leisure Decision Model

- Of the **24-hours a day** available to us, if we allocate 8 hours for biologically and culturally determined activities like sleeping, eating and others, the remaining 16 hours will be at our disposal.
- We can spend them either for work or for leisure, or some combinations of them.
- To analyze the workers' behavior in allocating the discretionary hours between work and leisure, economist employ the **neoclassical work-leisure decision model**.
- Having the goal of identifying the factors that determine whether to work at all and, if so, how long to work, the model helps us
 - understand the facts about labor supply
 - predict the influence of complex policy issues on the labor supply behavior.
- The model regards workers as having the **objective of utility maximization** in which preferences of a worker and budget constraints are key concepts.

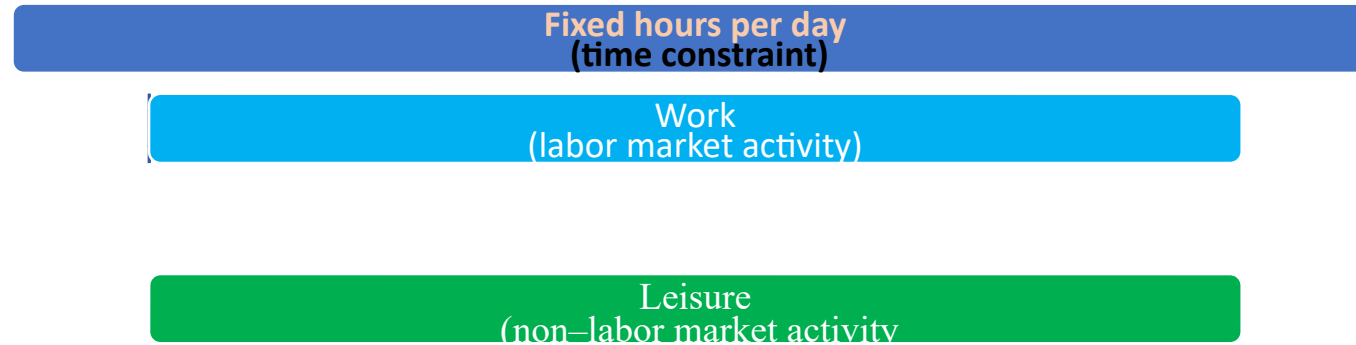
2.2. The Work-Leisure Decision Model

- The **representative person** in our model **gets satisfaction** both from the consumption of goods & services (denoted by **C**) and from the consumption of **leisure time (L)**.
- **Q#1**. How we aggregate food, cloth, car, etc. to denote all by C?
- The notion that individuals get satisfaction from consuming goods and leisure is summarized by the utility function:

$$U = f(C, L)$$

- The utility function transforms the person's consumption of goods and leisure into an index U (called utility) that measures the individual's level of satisfaction or happiness.
- The higher the level of index U, the happier the person.
- We make the sensible assumption that buying more goods or having more leisure hours both increase the person's utility, i.e., **C & L are normal good**.

2.2. The Work-Leisure Decision Model

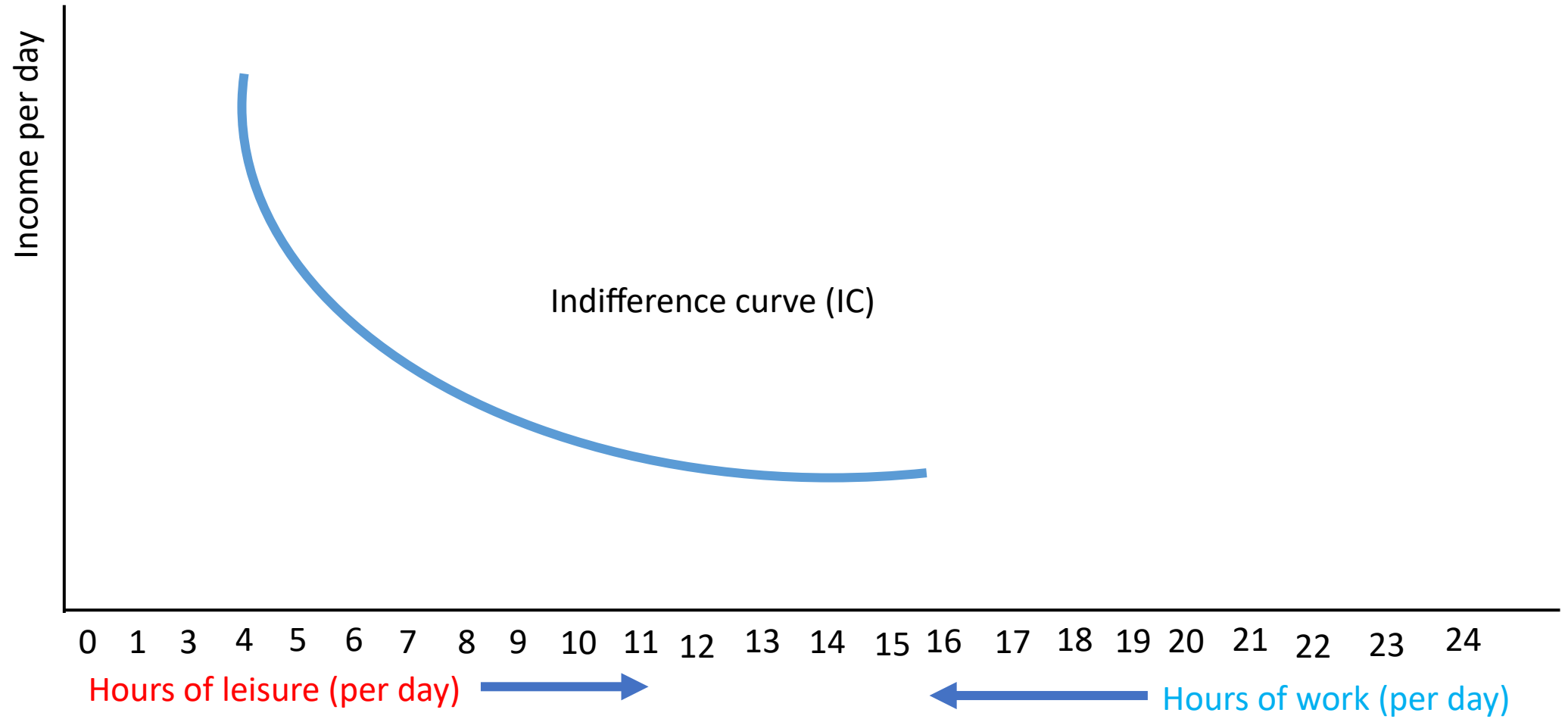


- **Work** is time devoted to a paying job.
- **Leisure** includes all kinds of activities for which a person does not get paid: time spent on consumption, education, commuting, rest, relaxation, and so forth.
- **Two sets of information** are necessary to determine the optimal distribution of an individual's time between work and leisure.
- **First**, we require **subjective, psychological information** concerning the individual's work–leisure preferences. This information is embodied in **indifference curves**.
- **Second**, we need the **objective market information** that is reflected in a **budget constraint**.

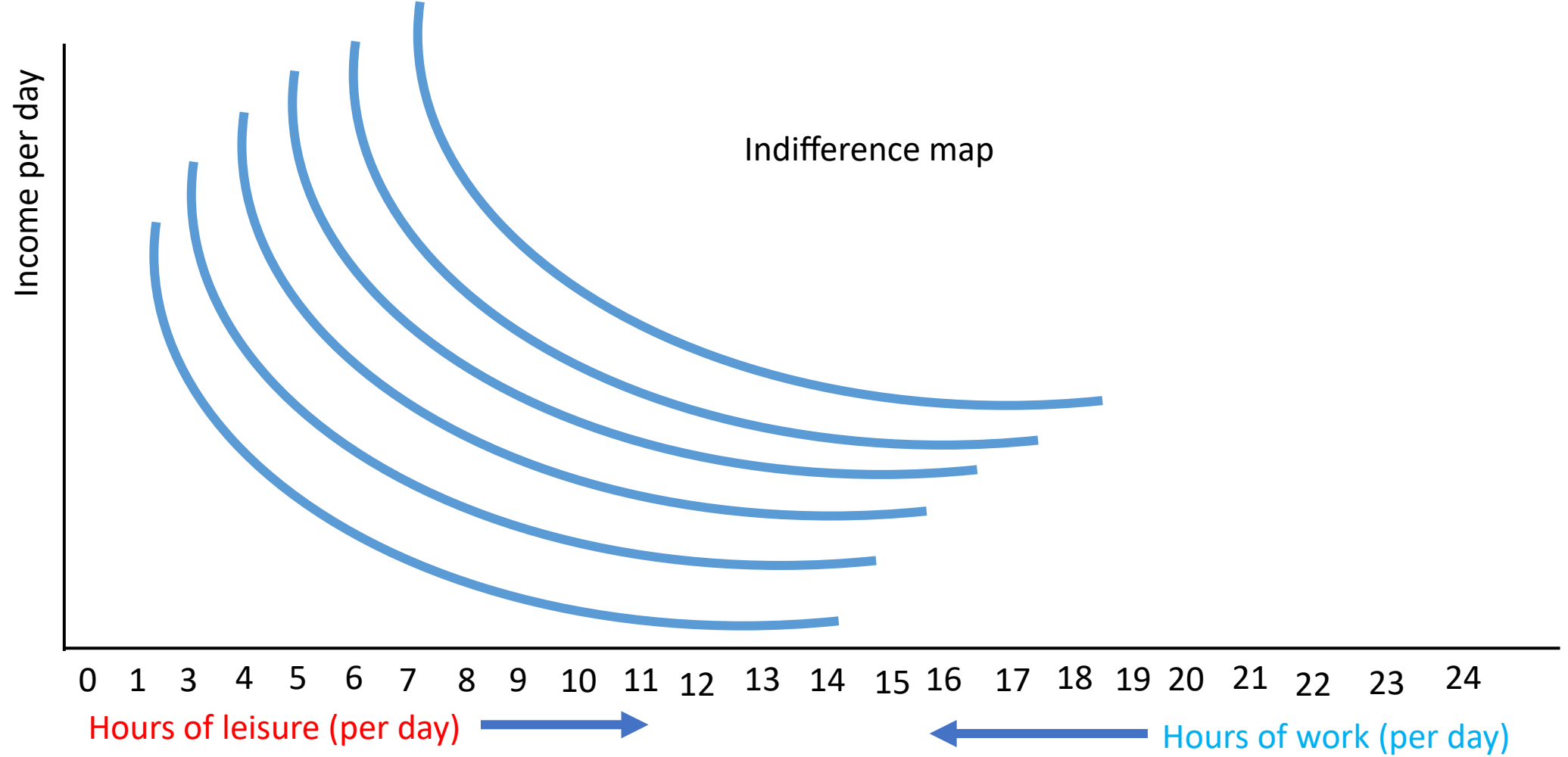
2.2. The Work-Leisure Decision Model

- C and L, are to some extent **substitutes** for each other.
- ➔ various combinations of C and L can provide identical level of satisfaction - if a worker is made to forgo some hours of leisure for work, s/he would enjoy an increment on his money income so that his/her satisfaction could keep unchanged.
- The locus of points that stand for equal utility is termed as an **indifference curve**.
- We can construct an indifference curve for every level of utility ➔ indifference map.

2.2. The Work-Leisure Decision Model



2.2. The Work-Leisure Decision Model



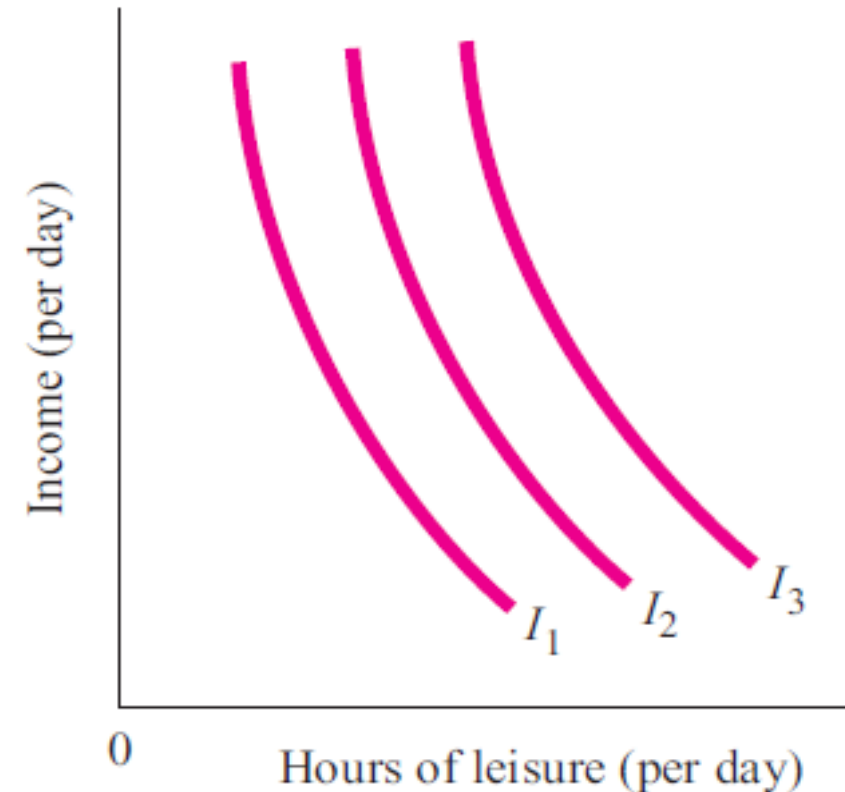
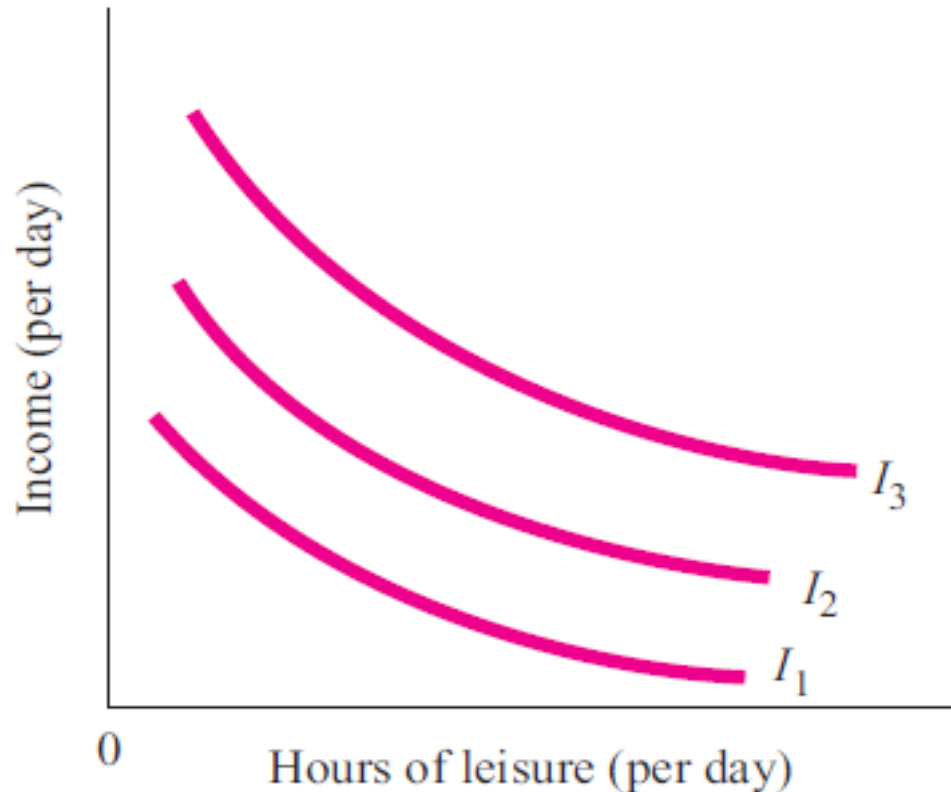
2.2. The Work-Leisure Decision Model

Properties of indifference curves:

1. Indifference curves are downward sloping. More L and/or C means nothing otherwise → contradicts the assumption that L and C are normal goods.
2. Indifference curves do not intersect. Why?
3. Indifference curves are convex. Why? W/c good you would like to give up more to get another good: the good that you have large amount or little?
4. Higher indifference curves indicate higher levels of utility

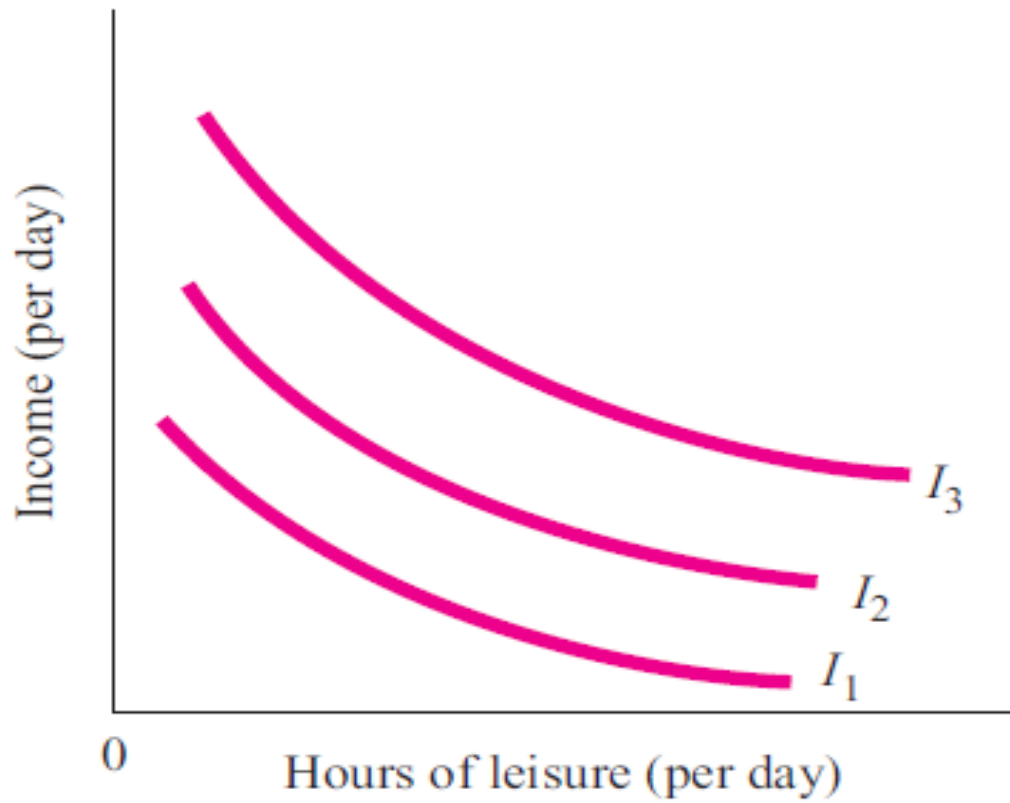
2.2. The Work-Leisure Decision Model

Different Preferences for Work (Income) and Leisure: w/c one depicts 'leisure lover'. How do you know this?

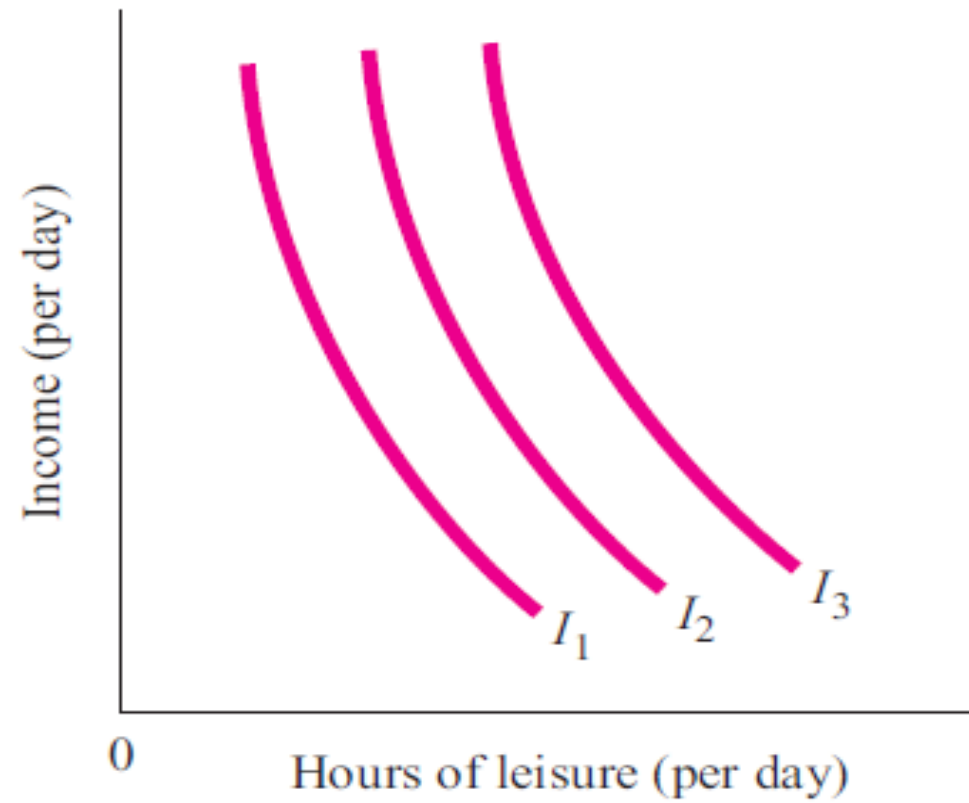


2.2. The Work-Leisure Decision Model

Different Preferences for Work (Income) and Leisure: w/c one depicts ‘leisure lover’. How do you know this?



(a) A “workaholic”



(b) A “leisure lover”

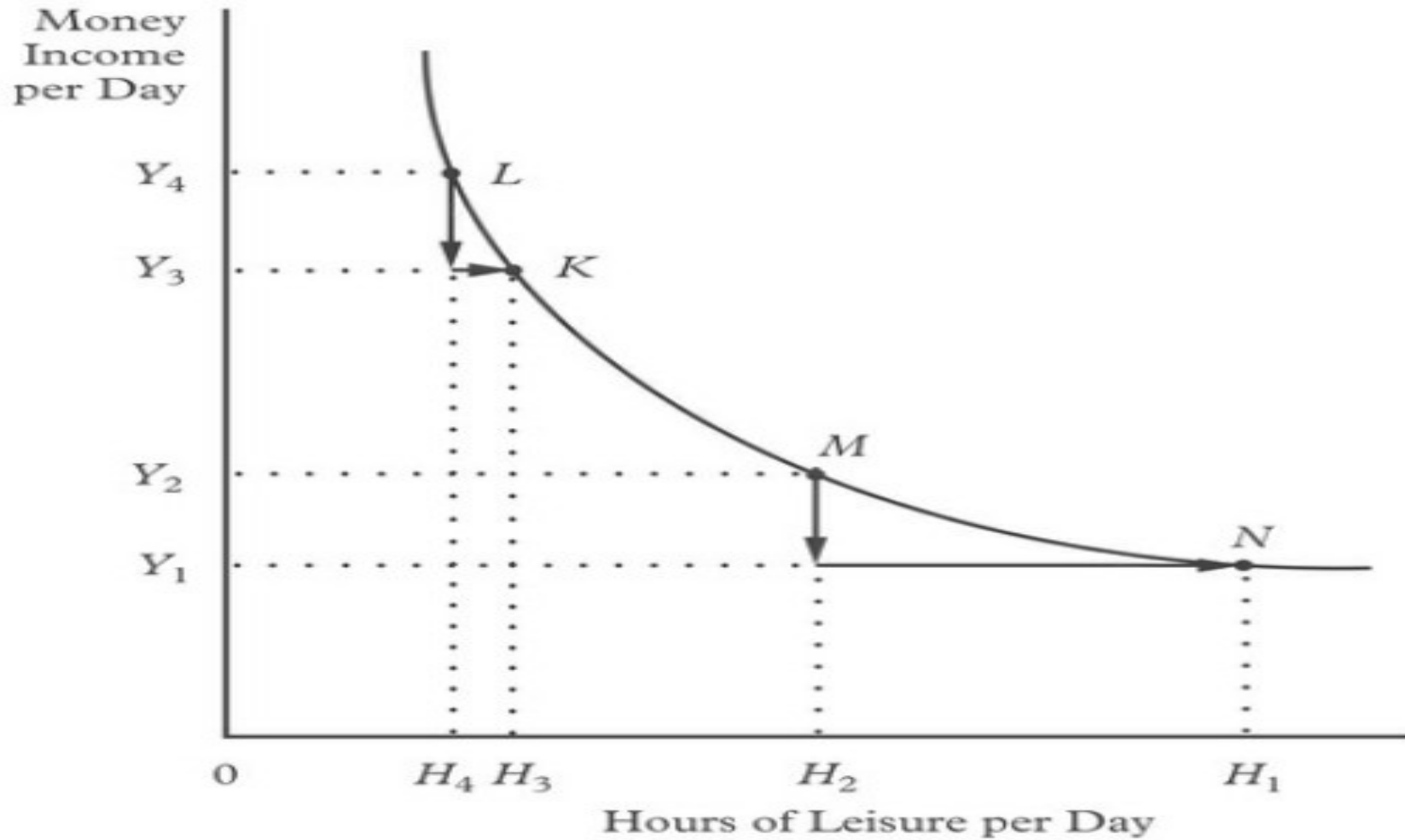
Slope of the IC

- Measures the rate at which a person is willing to give up some leisure time in return for additional consumption, while holding utility constant.
- The slope tells us how many additional dollars' worth of goods it would take to induce the person into giving up some leisure time.
- A movement from east to west along the indifference curve reduces utility by the product of the number of leisure hours being given up, ΔL and the associated loss of marginal utility MU_L .
- The Forgone leisure hour is used to work and earn additional income to be used for consumption, w/c increases utility by the product of ΔC , and the marginal utility derived from the consumption of additional good purchased by using the additional money income MU_C .
- Since the movement along the indifference curve involves no change in total utility, the loss is exactly compensated by the gain.

Slope of the IC

- That is,
- $\Delta u = \Delta f(C, L) = 0$
 $(\Delta L \times MU_L) + (\Delta C \times MU_C) = 0$
- Upon rearranging, we have
- Slope of IC = $\frac{\Delta C}{\Delta L} = -\frac{MU_L}{MU_C}$
- The absolute value of the slope of an indifference curve, which is also called the **marginal rate of substitution (MRS) in consumption**, is the ratio of marginal utilities.
- Convexity implies that the slope of an indifference curve is steeper when the worker is consuming a lot of goods and little leisure, and that the curve is flatter when the worker is consuming few goods and a lot of leisure → **diminishing MRS**.

Slope of the IC



The budget constraint

- If the resources were not scarce, every worker would like to enjoy the ideally highest utility.
- But the discretionary hours available for leisure and work are limited.
- The resource that is under the command of the worker is constituted from the money income and leisure time.
- The money income may be obtained from various sources including
 - A. non-labor income** such as property rent, dividends , interest, remittance, profit and lottery
 - B. labor income** – employment compensation
- **Assume**
 - that of the total available daily time, T , the person allots h hours of for work
 - Wage rate is independent of hours worked.
- So, the person earns wh amount of income from work, where w is wage rate per unit of time, e.g., hour/week/month
- Thus, the person's budget constraint can be
- $C = wh + V$, where V is income from non-labor sources

The budget constraint (BC)

- The total time allocated to each of activities must equal the total time available in the period:

$$T = h + L \rightarrow h = T - L; \text{ where } L \text{ is leisure time}$$

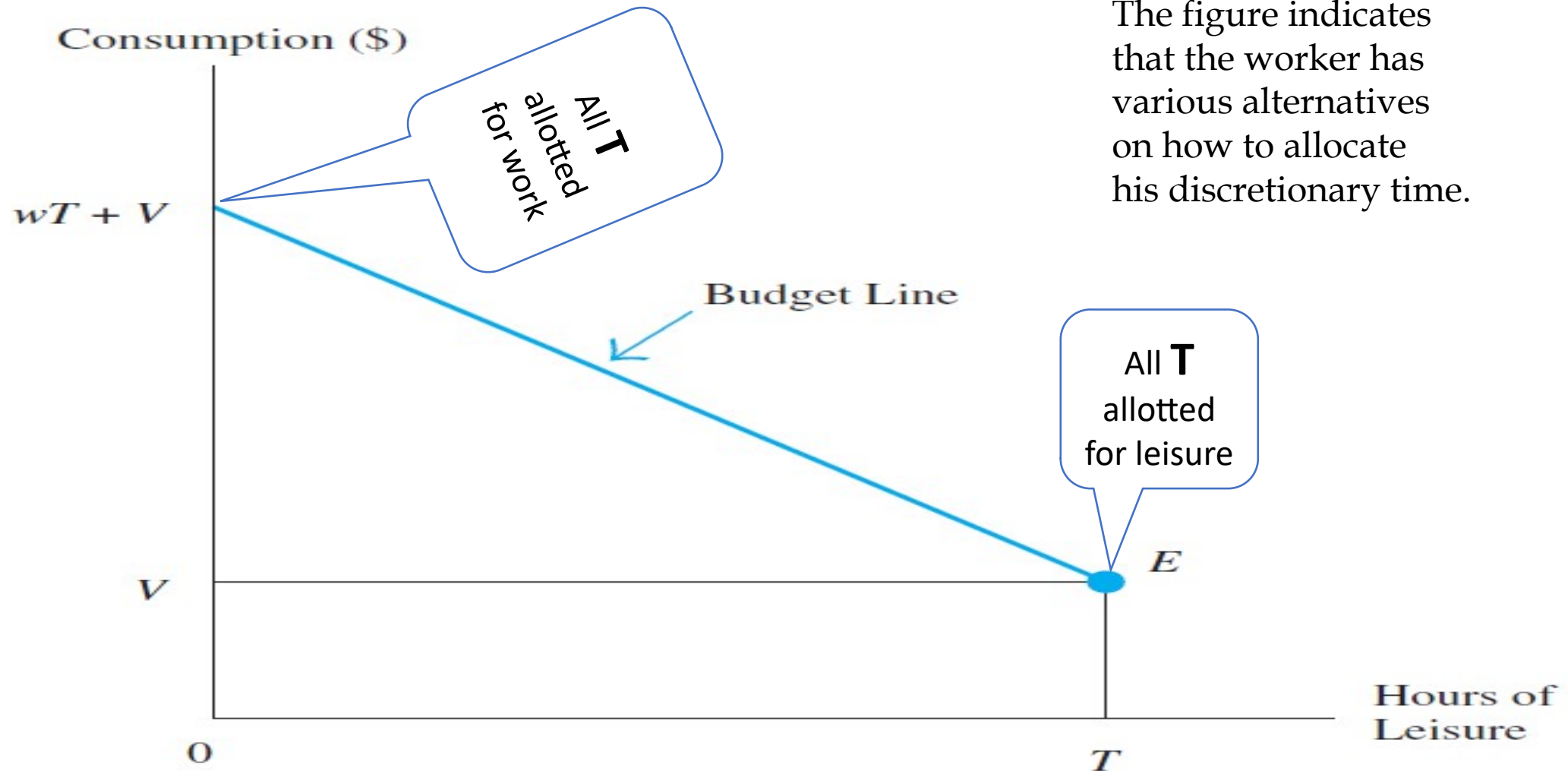
- The BC can be re-written as

$$C = w(T - L) + V \rightarrow C = (wT + V) - wL$$

- Slope = $-w$. Show.
- Draw the BL.

The budget constraint (BC)

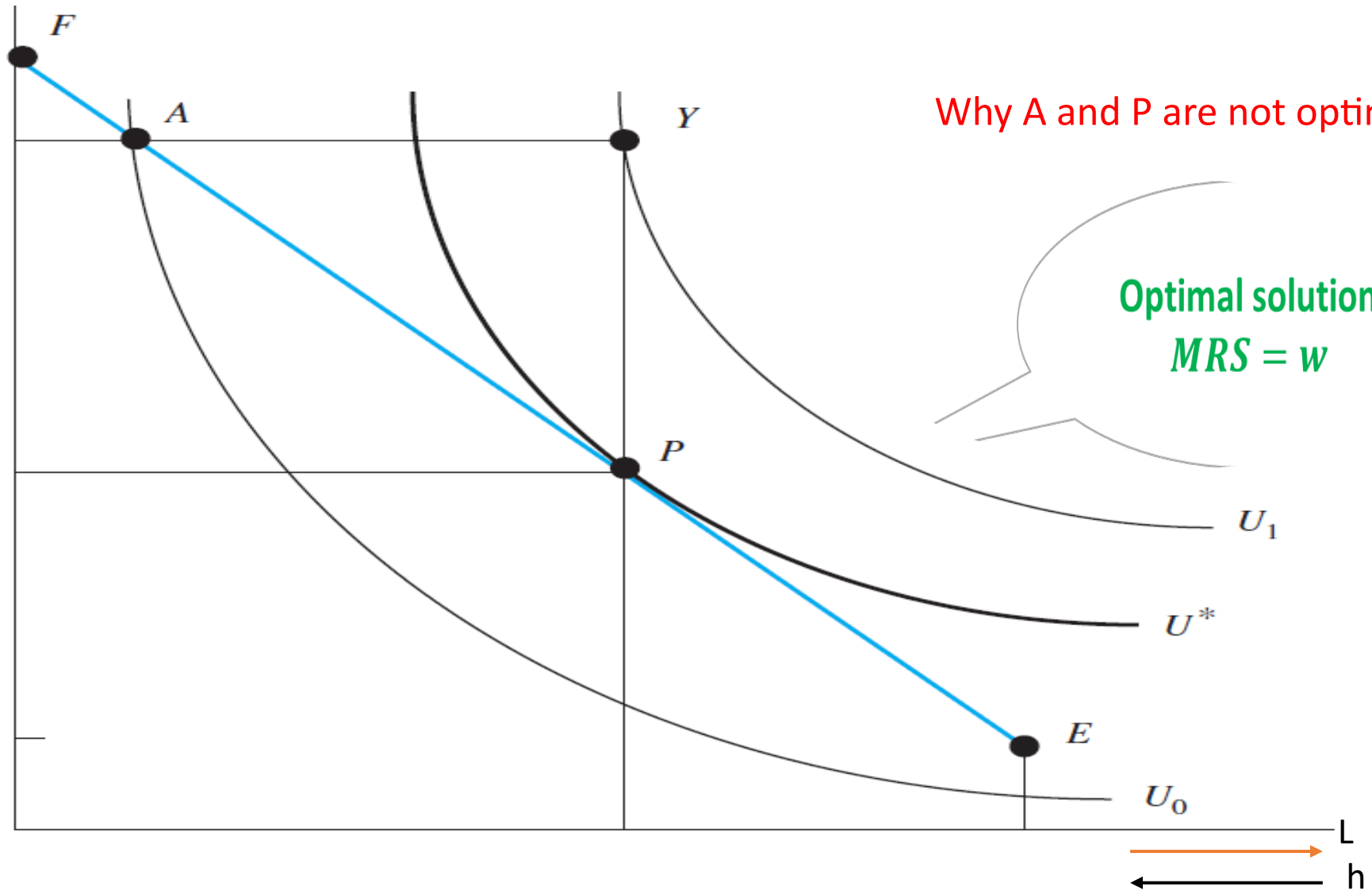
- Draw the BL.



- The individual's optimal or utility-maximizing position can be determined by bringing together
 - The **subjective** preferences embodied in the indifference curves and
 - The **objective market** information contained in the budget line.
- Recall that the farther the indifference curve is from the origin, the greater the person's total utility.
- Therefore, given the budget constraint, an individual will maximize total utility by attaining the highest possible indifference curve
- The **optimal** work–leisure position is achieved where $MRS_{L, C}$ (the slope of the indifference curve) is equal to the wage rate (the slope of the budget line).
- By definition, these slopes are equal only at the point of tangency.
- Implying that
 - the rate at which a **person is willing** to give up leisure hours in exchange for additional consumption (i.e., MRS) equals the rate at which the **market allows** the worker to substitute one hour of leisure time for consumption (the wage rate).
 - the last dollar spent on leisure activities buys the same number of utils as the last dollar spent on consumption goods.

The Hours of Work Decision (Utility maximization)

C



Exercise

- Suppose that a person earns \$1 per hour from working as a manager at a firm. Her utility function from consumption of goods and leisure is given by

$$U(C, L) = C^{0.5} L^{0.5}$$

- The person must sleep 8 hours of the day.
 1. Draw the BL
 2. Find the utility maximizing allocation of hours of a day.

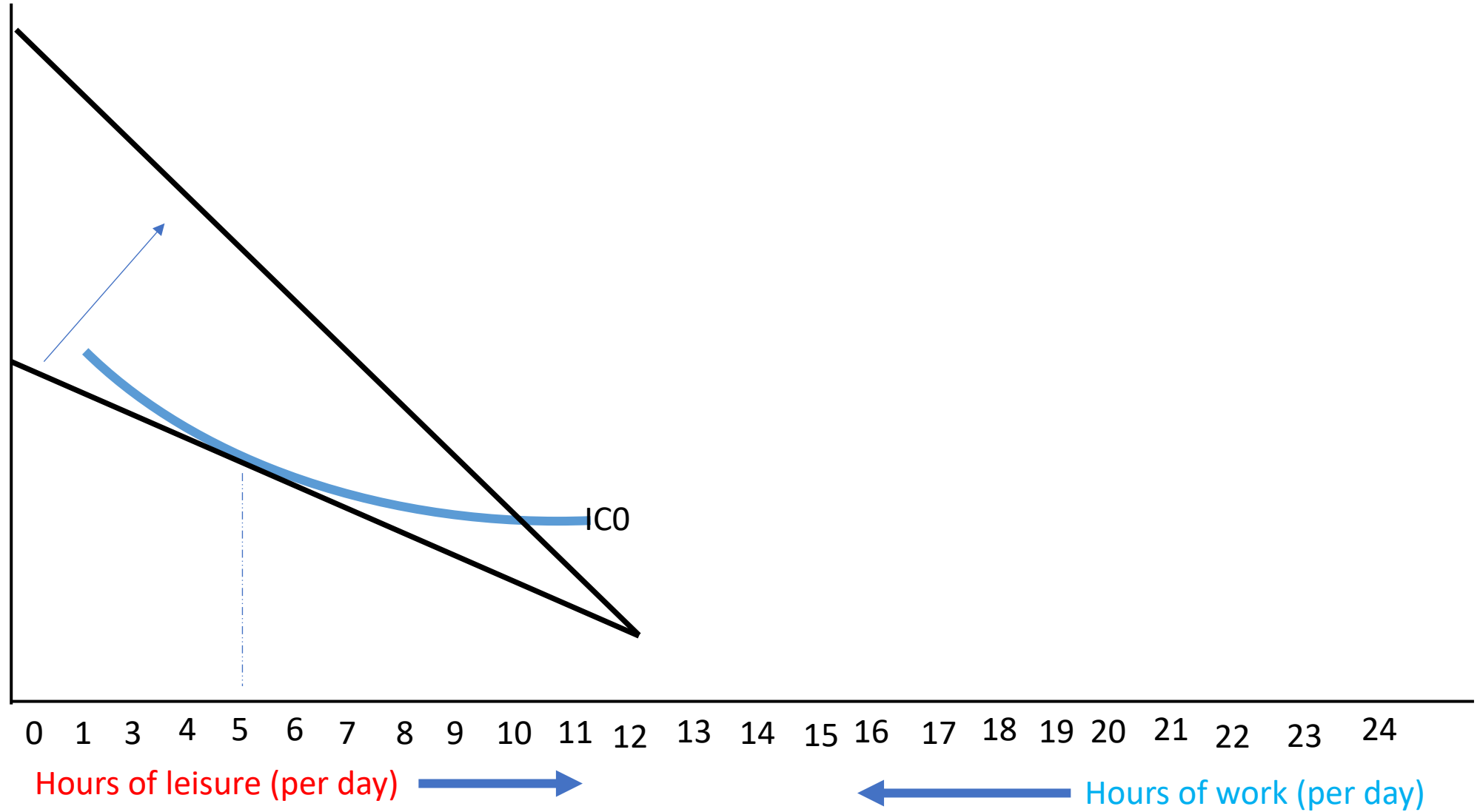
Scenarios: Change in wage rate

- Suppose that the **wage rate increases**.
- # Implications
 - A. Leisure time becomes expensive – the person wishes to **substitute** some of leisure hours for work. → an increase in wage rate induces substitution effect that increases working hours at the expense of leisure hours.
 - B. Income increases as the increased wage rate applied to all worked hours. The person becomes **wealthier**, and wishes to allot more hours for leisure, reducing working hours. → an increase in wage rate triggers income effect that increases leisure hours at the expense of working hours.
 - Net effect?
 - ✓ Opportunity set of the person increases, i.e., utility increases
 - ✓ On hours of work depends on the whether SE or IE dominates: an IE pushing toward less work while a SE pushing toward more work
- Which effect dominates on hours of work? Depends, among others, on
 - The initial income level
 - The prefers for leisure Vs money
 - external (to the person) factors, e.g., institutional factors

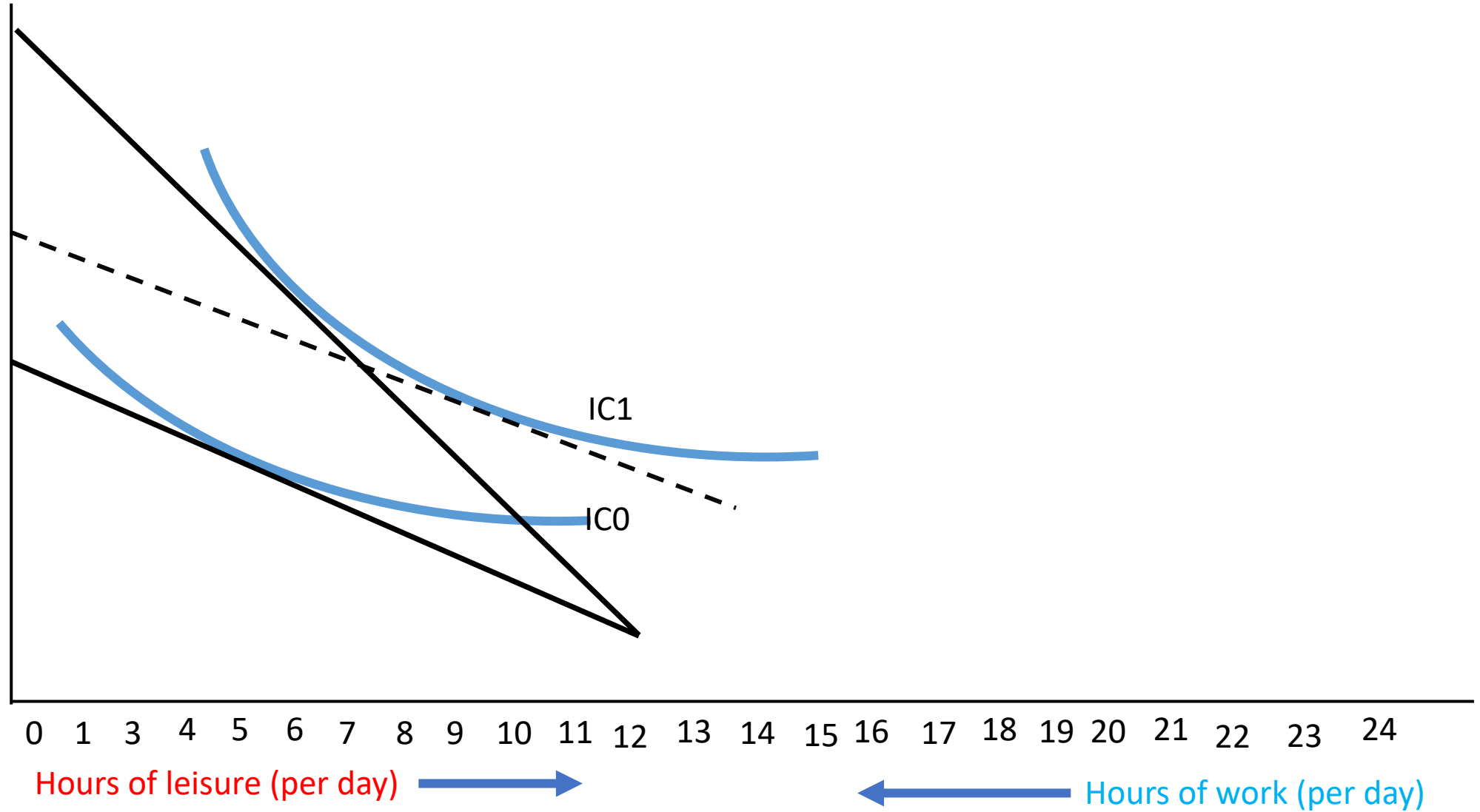
Scenarios: Change in wage rate

- Graphical illustration of wage rate increases
- Suppose that the person is **leisure lover** - steep IC
- To show the income effect, assume that the new BL is achieved from non-labor source where wage rate remains unchanged as the original wage rate.
- To show the substitution effect, consider the new IC w/c can be achieved from two differently sloped BLs: the new BL & another BL drawn parallel to the original BL.

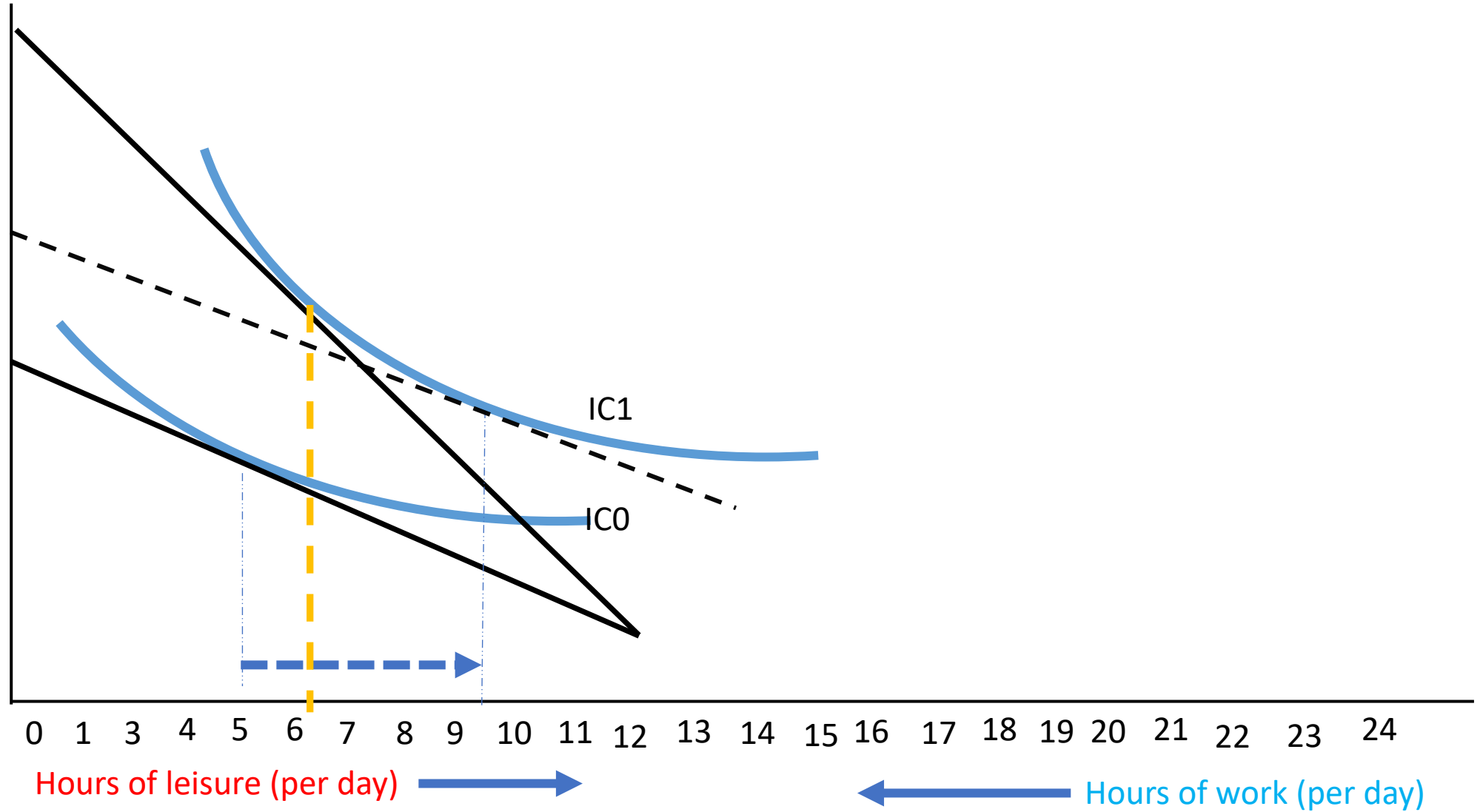
Scenarios: Change in wage rate



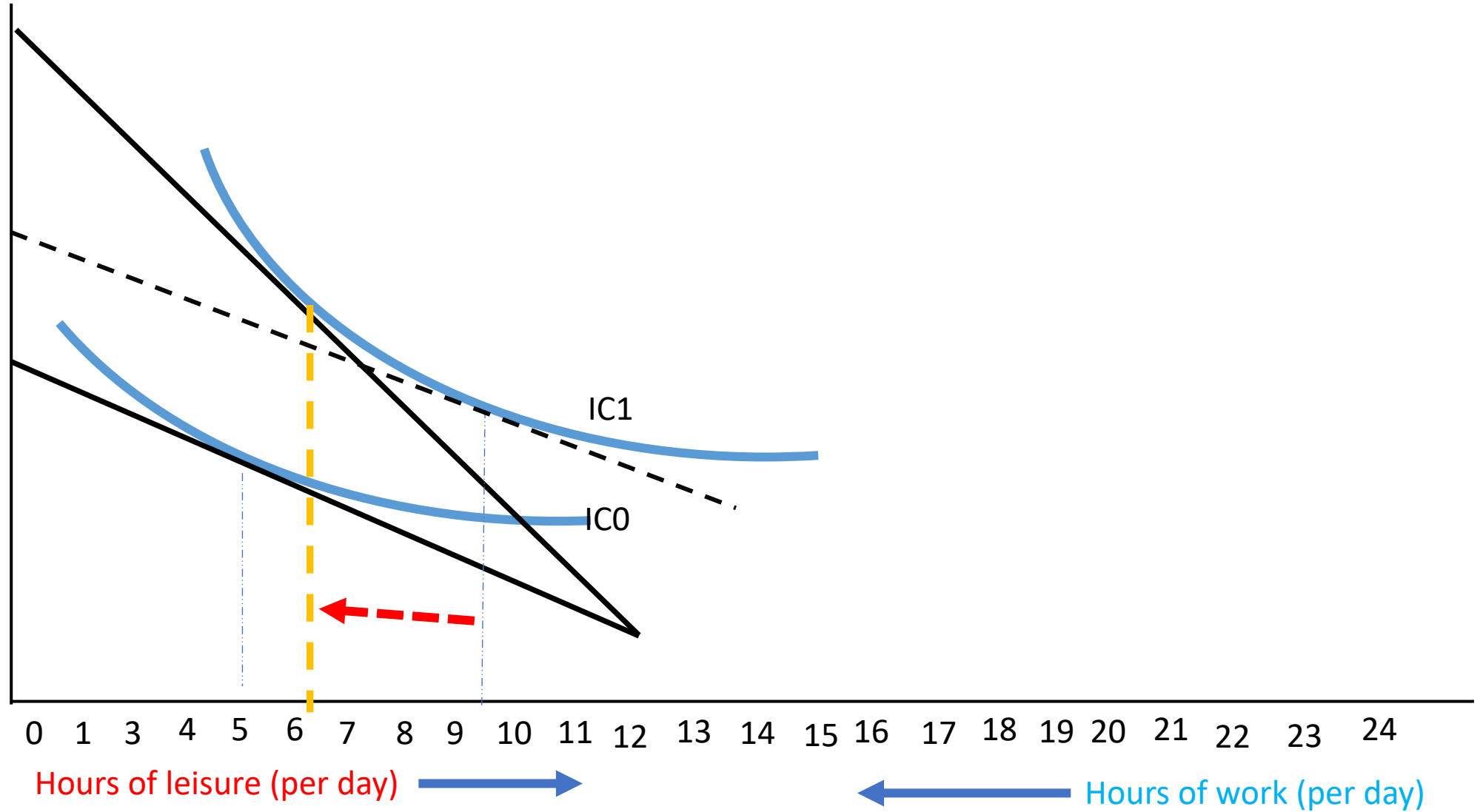
Scenarios: Change in wage rate



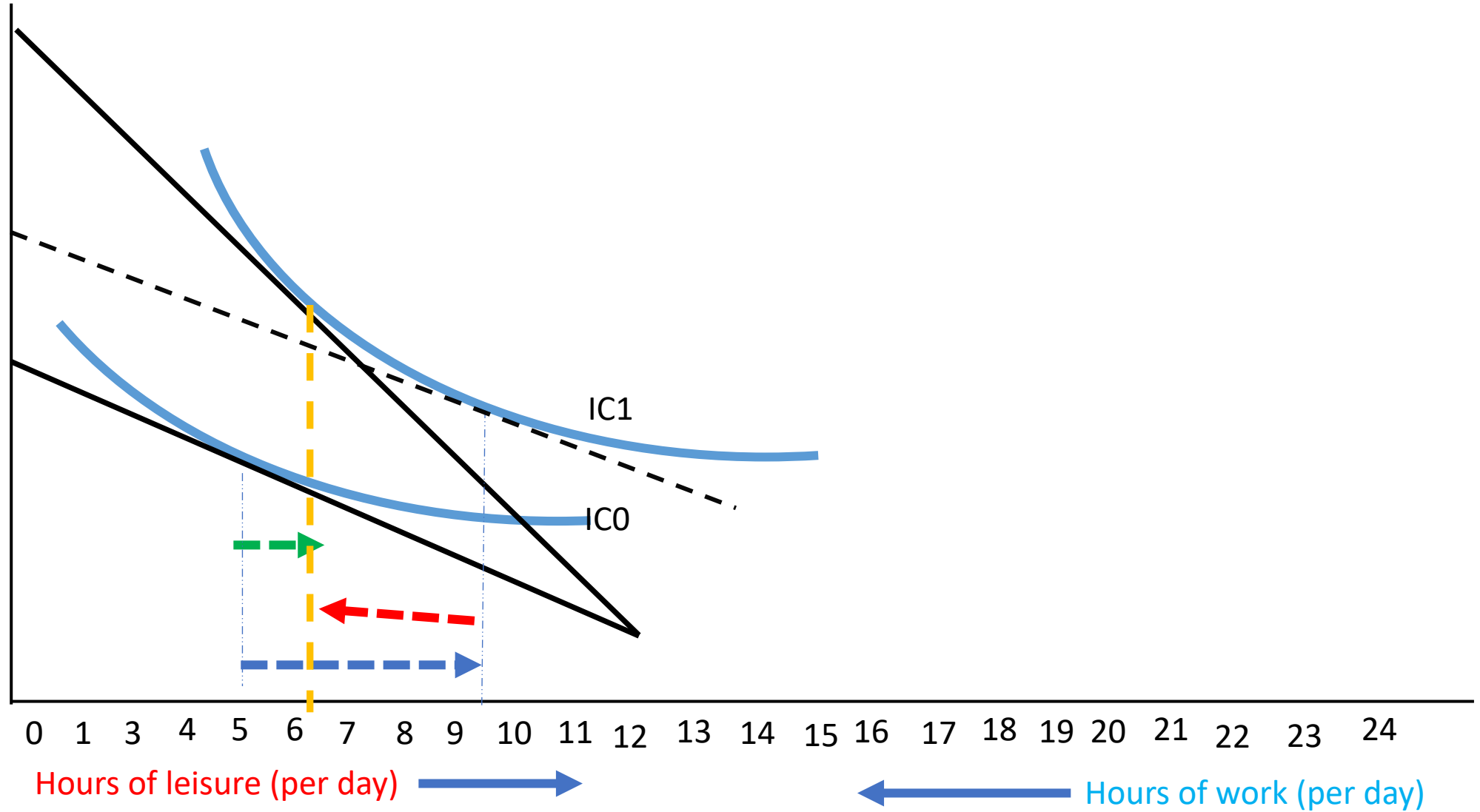
Scenarios: Change in wage rate



Scenarios: Change in wage rate



Scenarios: Change in wage rate



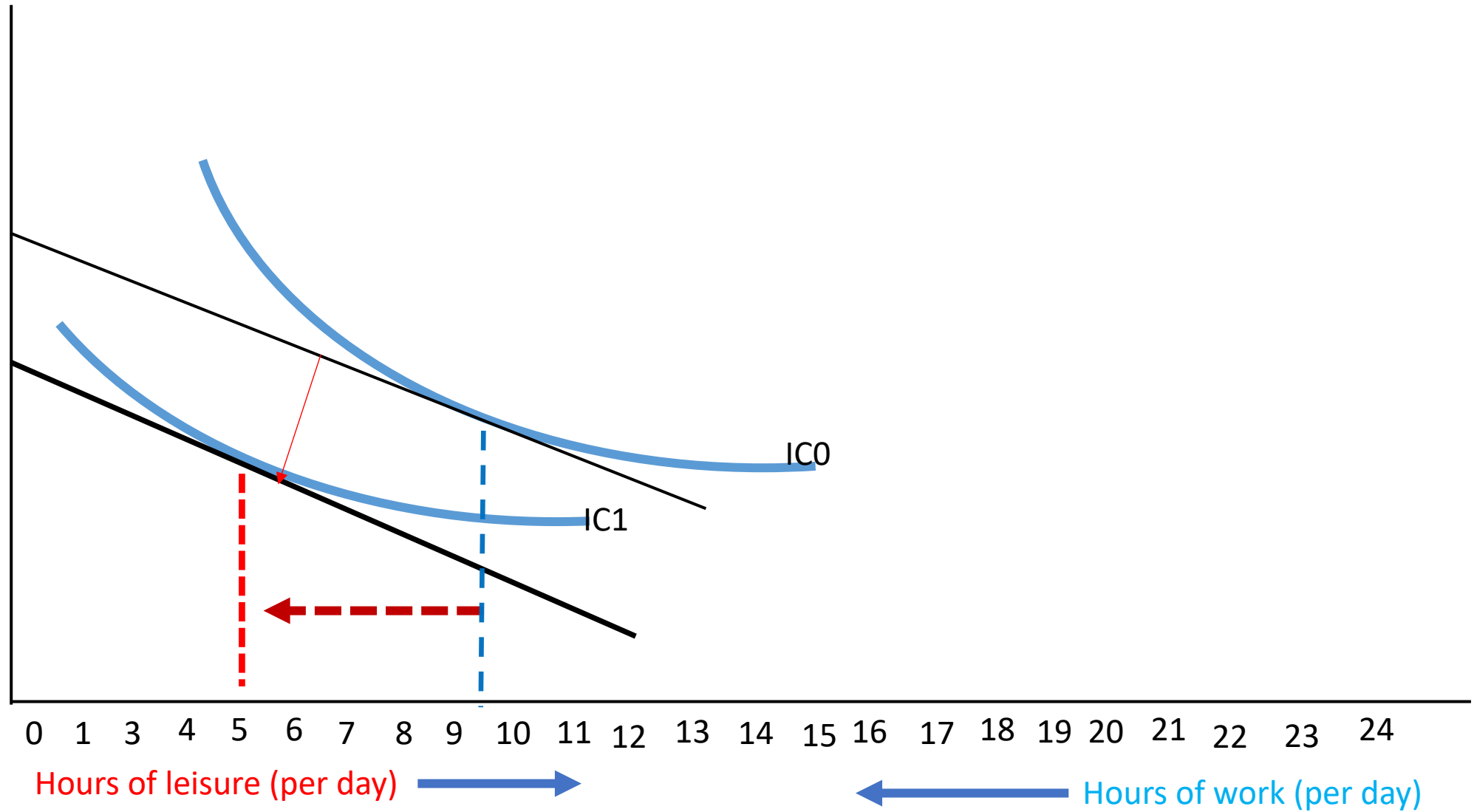
Scenarios: What Happens to Hours of Work When Nonlabor Income Changes?

- What is the impact on:
 - Opportunity sets (utility)
 - Hours of leisure
 - SE? IE?

Scenarios: What Happens to Hours of Work When Nonlabor Income Changes?

- Graphical illustration the effect of a loss in nonlabor income source
- The relative price of leisure (to work) remains unchanged as wage is unaffected.
- No SE
- Income effect
- The person becomes poorer now
- Work hours increase; leisure time decrease
- Utility decrease

Scenarios: Decrease in nonlabor income



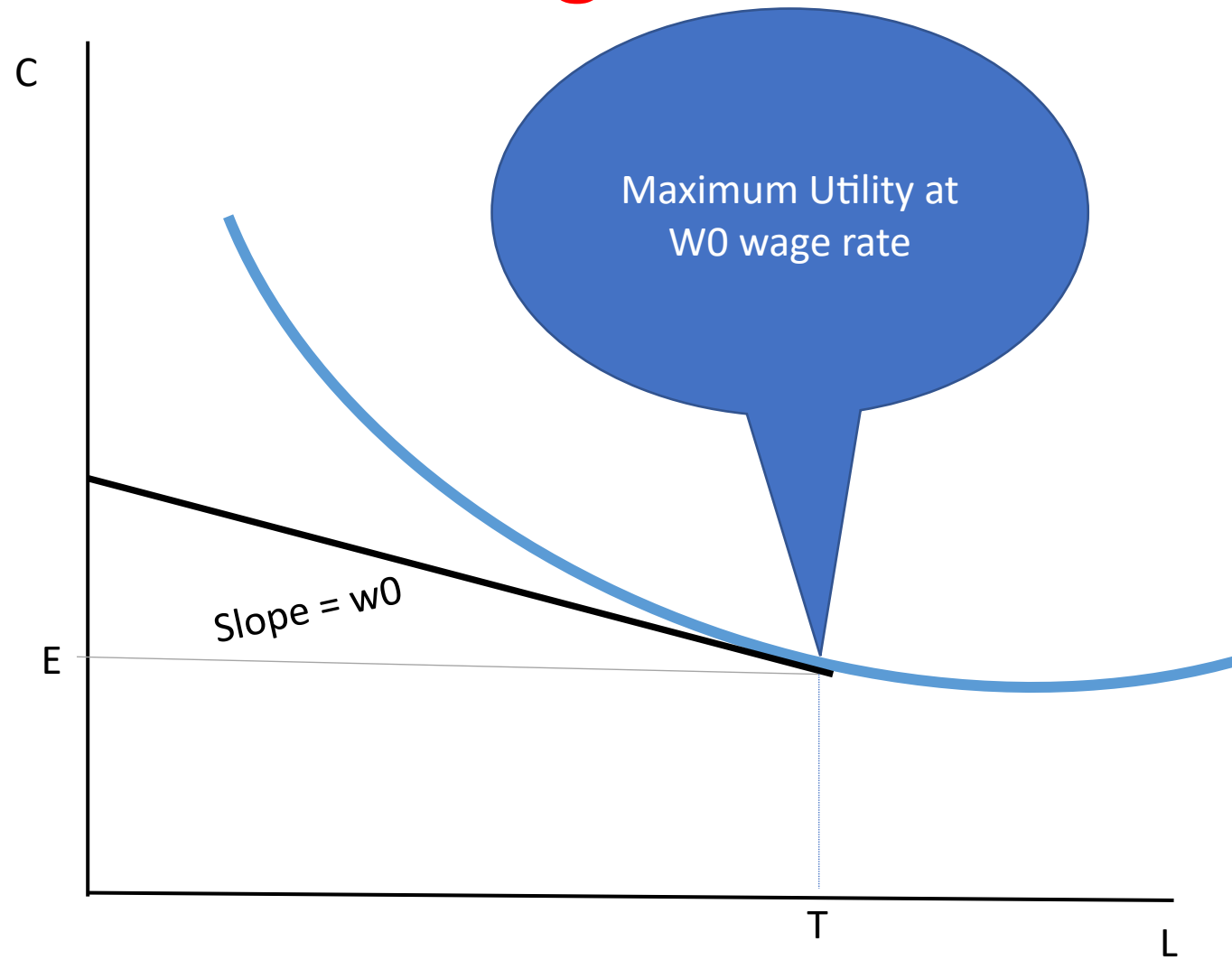
Exercise

- Graphically show the effect on (A) total utility and (B) work hours of a wage rate decrease when
 - i. IE dominates SE
 - ii. SE dominates IE.

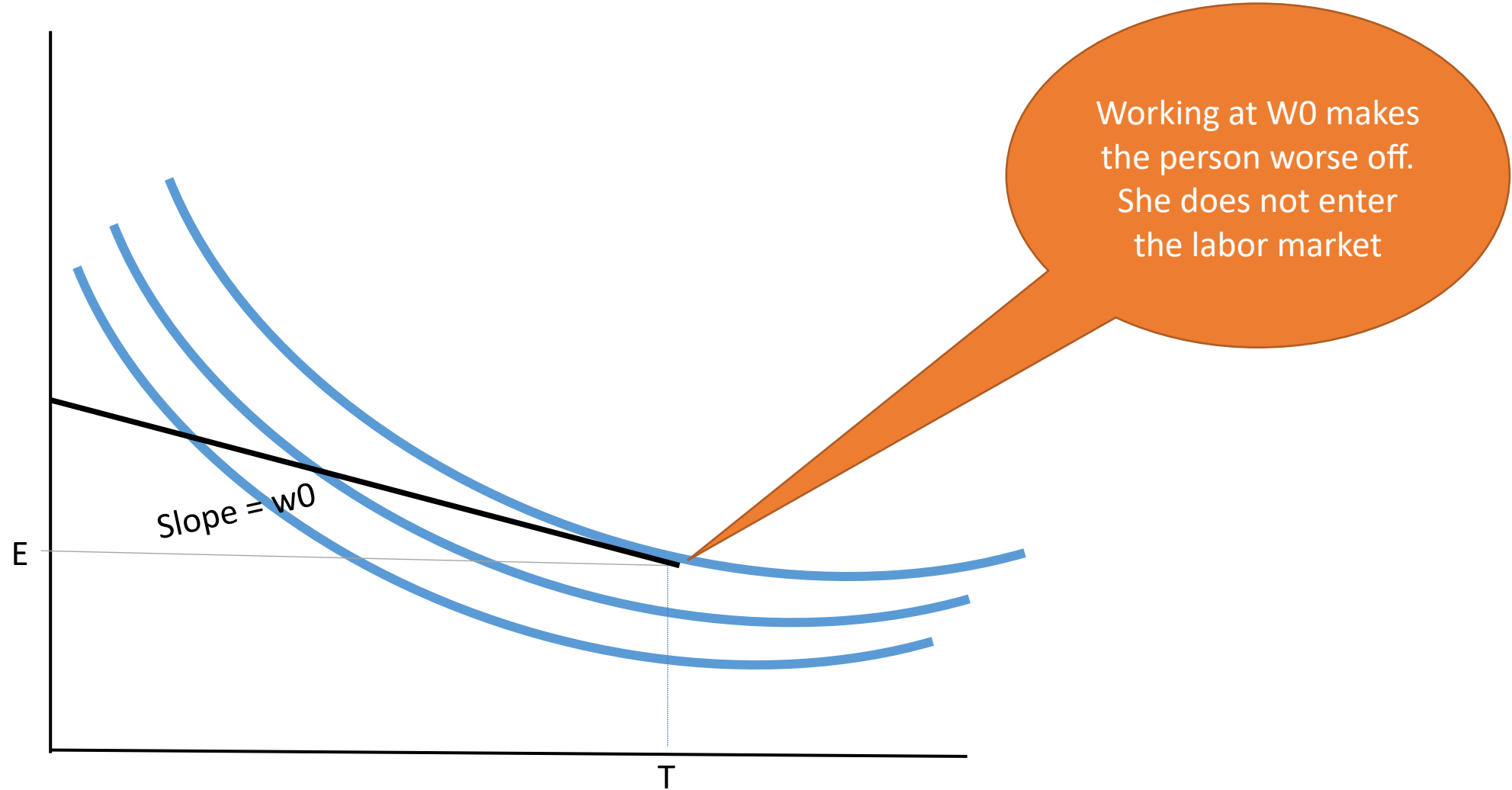
Reservation wage: to work or not to work

- What factors motivate a person to enter the labor force in the first place?
- One of the main factors affecting the decision to whether to work is **wage rate** while availability of alternative source of income and unemployment insurance plays also key role
- **Reservation wage**: the market wage at which the worker becomes indifferent between taking a job or not. It is the lowest wage rate at which one would decide to work.
- A worker places a value on the marginal hour of leisure time. If the offered wage rate is greater than this value, the worker will be willing to enter the labor force; otherwise not.

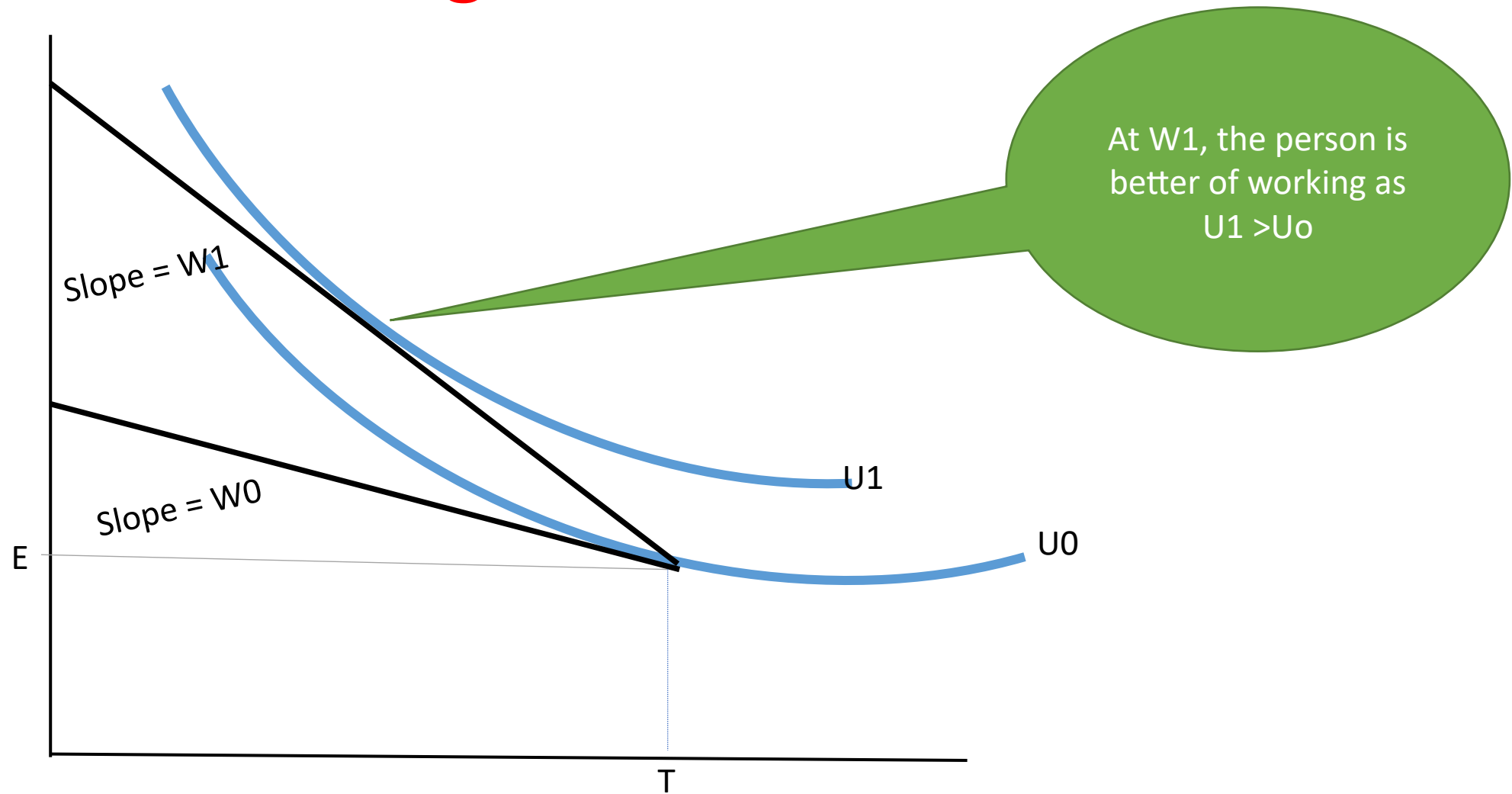
Reservation wage



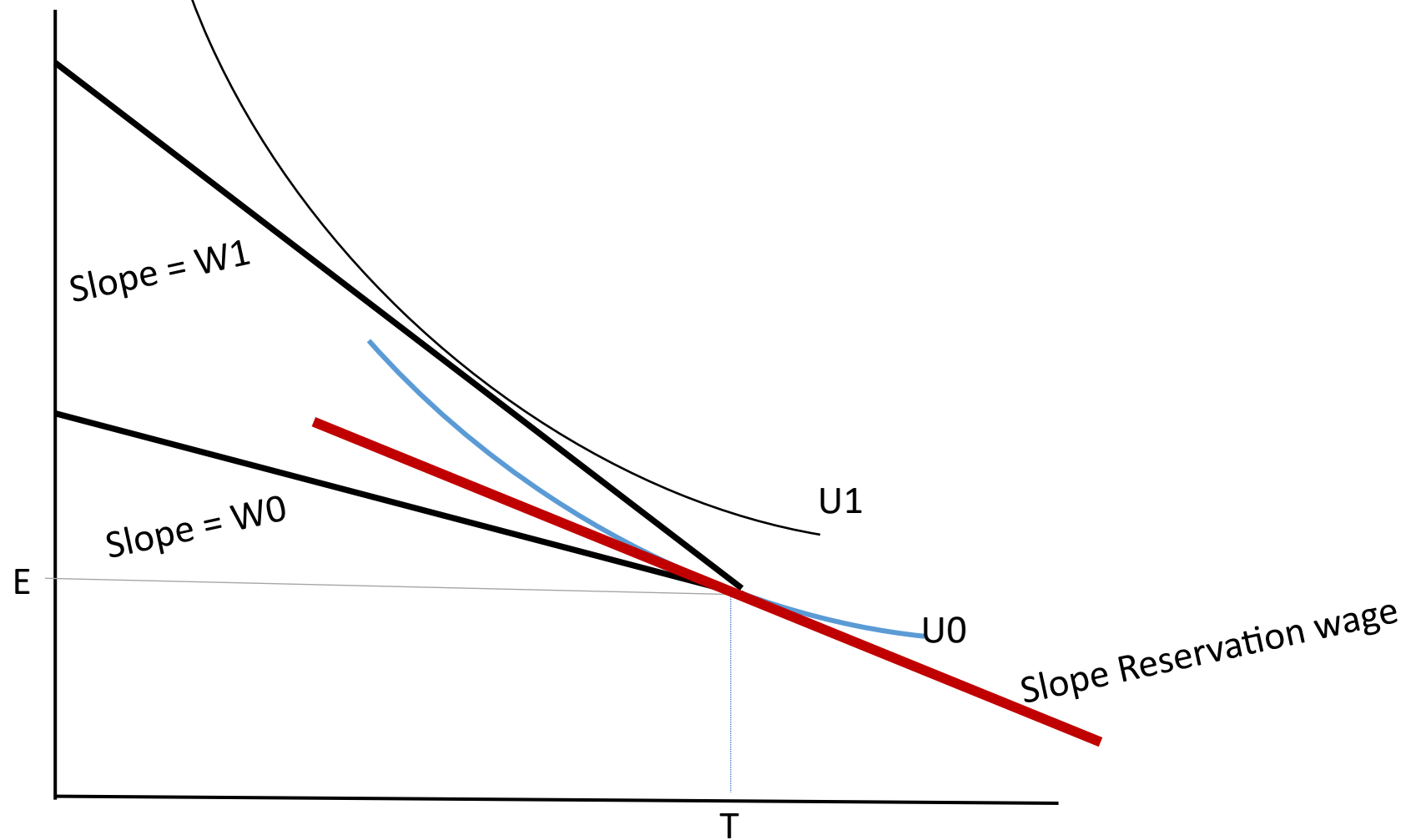
Reservation wage



Reservation wage



Reservation wage



Reservation wage

- The reservation wage gives the minimum increase in income that would make the person indifferent between remaining working and not working.
- Some implications of the reservation wage:
 1. The decision to take a job is made by comparing the reservation wage and the market wage. If the market wage is greater (less) than the marginal value of leisure time, the worker will (will not) participate in the labor market.
 2. Since the reservation wage depends on the person's taste for work, as the worker's non labor income increases, his/her reservation wage becomes higher and consequently the person is less likely to enter the labor force
 3. There is a positive relation between the market wage and the probability of working.
 4. An increase in the wage of a nonworker does not generate an income effect.
 5. The wage increase makes leisure time more expensive and hence is likely to draw the nonworker into the labor force.

2.3. The labor supply

- The reservation wage constitutes the starting point for labor supply decision.
- Once the market wage starts to increase above the reservation wage, the worker starts to allocate positive hours of work.
- Such a relation between the market wage rate and the number of hours allocated for work provides the **labor supply**.
- Labor supply:
 - Individual labor supply
 - Market labor supply
- We use the utility-maximization framework to derive **individual** labor supply curve
- The market labor supply is horizontal aggregation of individual LS at a given wage rate.

Individual labor supply

- Consider a continuous marginal increase in wage rate.
- At RW , $L_s = 0$, but as wage rate marginally increases, initially the SE dominates the IE and L_s increases. However, if the wage rate keeps on increasing, then $SE = IE$, and then $SE < IE$ and L_s declines.
- This gives **backward-bending labor supply curve** because it eventually bends around and has a negative slope in the bended segment
- **When is the L_s curve positively sloped?** When $SE > IE$ or $IE > SE$?
- To find the individual L_s , record the hours of work for each wage rate, then present the L_s by
 - Graph
 - Table
 - Equation

Determinants of Ls

- Factors that induce **shift** in Ls include
 1. Non-labor income
 2. Change in work-leisure preference
- Factor that induce **movement** along the LS
 - Wage rate
 - Income tax

Q#: Will LS shift to the left or right, and what happens to LS when:

1. A spouse fired from job?
2. Children school fees increase for families having children
3. Preference for leisure increases
4. Government decreases dividend tax for shareholders
5. An improvement in working conditions,
6. Availability of child care service

Exercise

- Q#2:

1. Graphically derive individual labor supply.
2. Given the following information
 1. Draw individual LS curves
 2. Draw the market labor supply
 3. Drive the slope of the market LS

Wage rate	1	2	3	4	5	6	7	8	9
Mr. X hours of work	0	0	0	2	3	5	7	8	7
Mr. Y hours of work	0	2	5	7	7	5	4	3	2

Wage Elasticity of Labor Supply

- The elasticity of labor supply measures the extent of responsiveness of hours of work to changes in the wage rate.
- Gives the percentage change in hours of work associated with a 1% change in the wage rate.
- $E_w^{ls} = \frac{\% \text{ change in hours of work}}{\% \text{ change in wage rate}} = \frac{\partial Ls}{\partial W} * \frac{W}{h}$
- Over specific ranges of an individual's labor supply curve, the elasticity coefficient could be
 - Zero (perfectly inelastic) – horizontal LS curve
 - Infinite (perfectly elastic) -
 - Less than 1 (relatively inelastic),
 - greater than 1 (relatively elastic),
 - or negative (backward-bending).

Wage Elasticity of Labor Supply

- The elasticity will depend on the relative strengths of the IE and SE.
 - $E_W^{ls} > 0$ if SE ----- IE?
 - $E_W^{ls} = 0$ if SE ----- IE?
 - $E_W^{ls} < 0$ if SE ----- IE?
- Example: suppose that a person works 8 hours per week at \$100 wage rate. A 10% imposition of tax induces the worker to reduce working hours by half. Find E_W^{ls} .

Empirical evidence

Sharp gender difference in wage elasticity of LS.

Women

Time allocation

Market work

Pure leisure

Work at home

Men

Time allocation

Market work

Pure leisure

Empirical evidence

- There is usually limited substitution b/n pure leisure time and hours of work for a working person.
- Wage rate increase does not significantly affect hours of working for men.
- Both SE and IE are small for men, where usually $SE \geq IE$
- A nearly vertical labor supply curve for men.
- The labor market participation rate for women is significantly less than that for men; many women work part-time, and women assume major responsibility for work within the home.

Empirical evidence

- For many married women, work in the home and work in the labor market are highly substitutable.
- When wage rates increase, many women substitute labor market work for work in the home.
 - Hire housecleaners
 - Hire babysitter or child care service
 - Purchasing prepared meals
- A strong SE, which implies an upward-sloping labor supply curve for married women.

Empirical evidence

- However, the gender gap in wage elasticity of LS is declining overtime since
 - A. Women market work participation is increasing overtime
 - B. Men and women more equally sharing home and market responsibilities
 - C. Technological changes such as stoves, washing machines, microwave, oven, etc. drastically reduce women time for home work activities,
 - ➔ freeing up time for leisure activities and for work in the labor market, and
 - ➔ reduce women specialization in home activities
 - D. Decline in fertility
 - E. Changes in cultural and legal attitudes toward working women
- Increase in the number of children raises a woman's **reservation wage** and reduces the probability that the woman will work.
- It is also likely, however, that the rise in the market wage made childbearing a very expensive household activity, reducing fertility – **reverse causality problem**.

Empirical evidence

- It has been estimated that about 60% of the total growth in the female labor force between 1890 and 1980 can be attributed to the rising real wage of women.
- **Husband wage Vs wife's Ls:** the labor force participation rates and hours of work of married women respond to changes in the husband's wage. A 10 percent increase in the husband's wage lowers the participation rate of women by 5.3 percentage points and reduces the hours that working wives allocate to the labor market by 1.7 percent.
- **Wife wage Vs husband' Ls:** little evidence
- Narrowing wage gap within the household further weakens the incentives for specialization (of women in home activities and men as bread-winner)

2.4. Policy Applications of the theory

- A) **The Labor Supply Implications of Welfare Programs.** Also called
 - income maintenance programs
 - public assistance programs
- **Purpose** of these programs – supporting the poor by providing **some minimum level of income** to all families and individuals.
- Income maintenance **programs should**
 1. Effectively get poor people out of poverty,
 2. maintain incentives to work, and
 3. Achieve goals 1 and 2 at a reasonable cost

2.4. Policy Applications

- **The Labor Supply Implications of Welfare Programs.** Also called
- Examples
 - Safety net program (e.g., Ethiopia)
 - Unemployment insurance
 - Aid to Families with Dependent Children (AFDC)
 - Temporary Assistance for Needy Families (TANF)

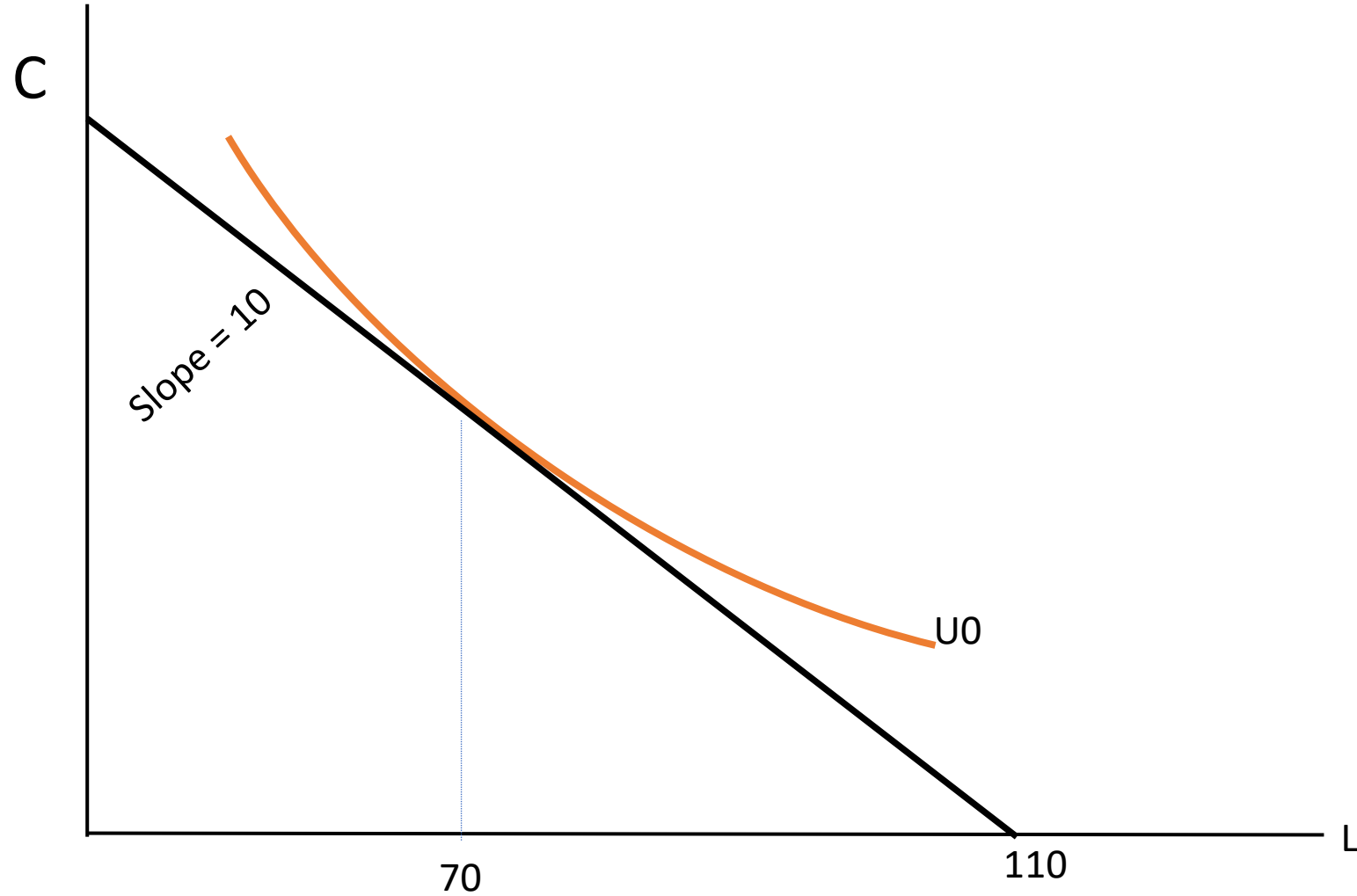
Q# What are the impacts on Ls of these programs?

A) The Labor Supply Implications of Welfare Programs

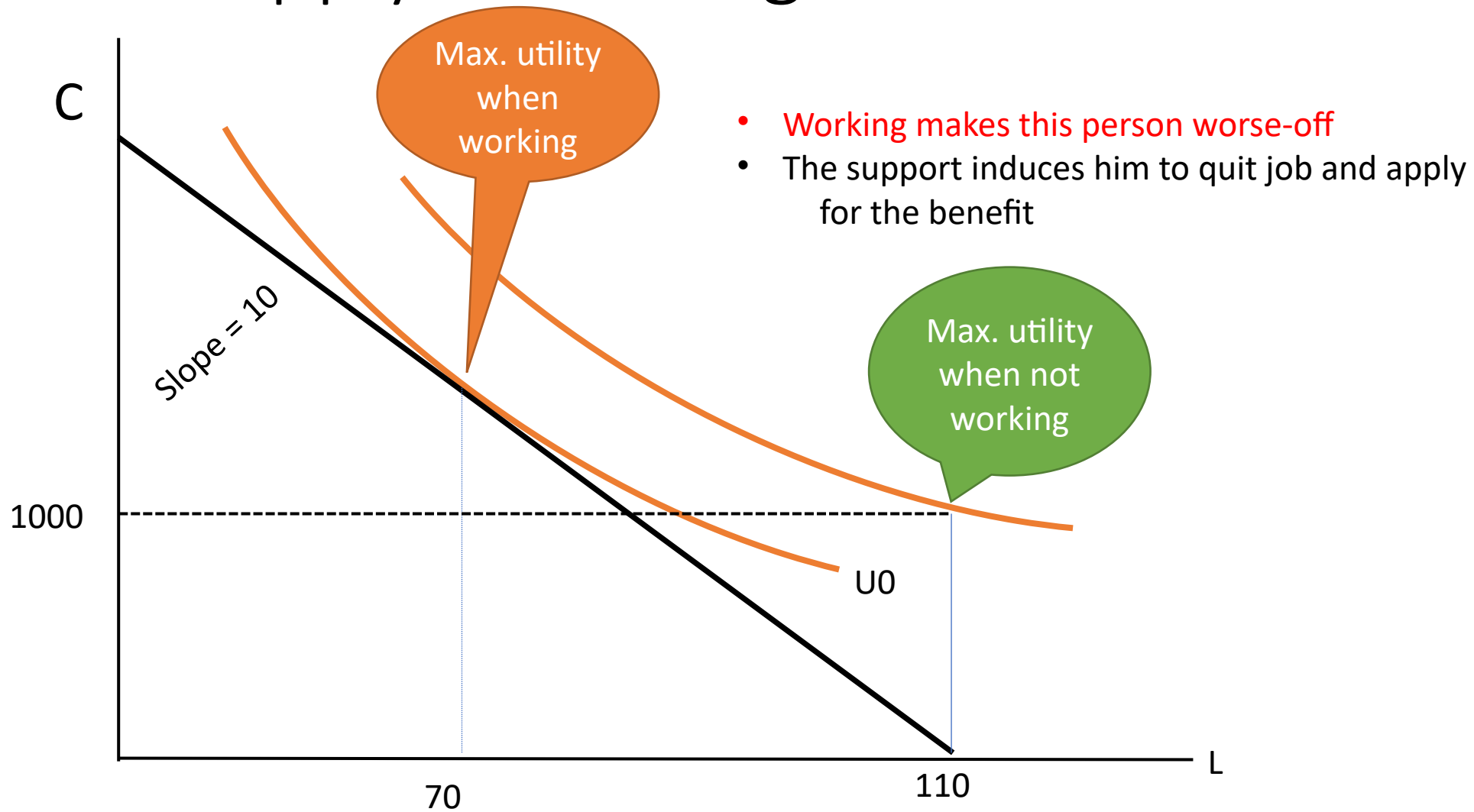
1. Consider a **take-it-or-leave-it cash grant** for eligible persons

- **Eligibility**: remain outside the labor force.
- If these persons enter the labor market, the government immediately drops the person from the welfare rolls (regardless of how much they earned from employment).
- Suppose that the person has no other source of income, and receives \$1,000 if not working and zero from the gov't if she works.
- Suppose also that the person works 40 hrs per week with \$10 per hour wage rate as shown in the graph below.
- **Q#** Assuming 110 discretionary weekly hours, show the impact of the grant on labor supply

Labor supply without the grant



Labor supply with the grant



Impact of take-it-or-leave-it cash grant

- The **grant increased the reservation wage rate**, reducing the likelihood that a low-wage person will enter the labor market.
- Thus, the **grant induce low-wage workers to drop out of the labor market**, not because these workers lack a “work ethic”, but b/c they are better off with the grant.
- This problem has been the main challenge in welfare programs.
- The **cash grant attracts low-wage earning workers to stop work** and receive the grant
- **Challenge**: how to support the poor with little or no effect on their Ls?

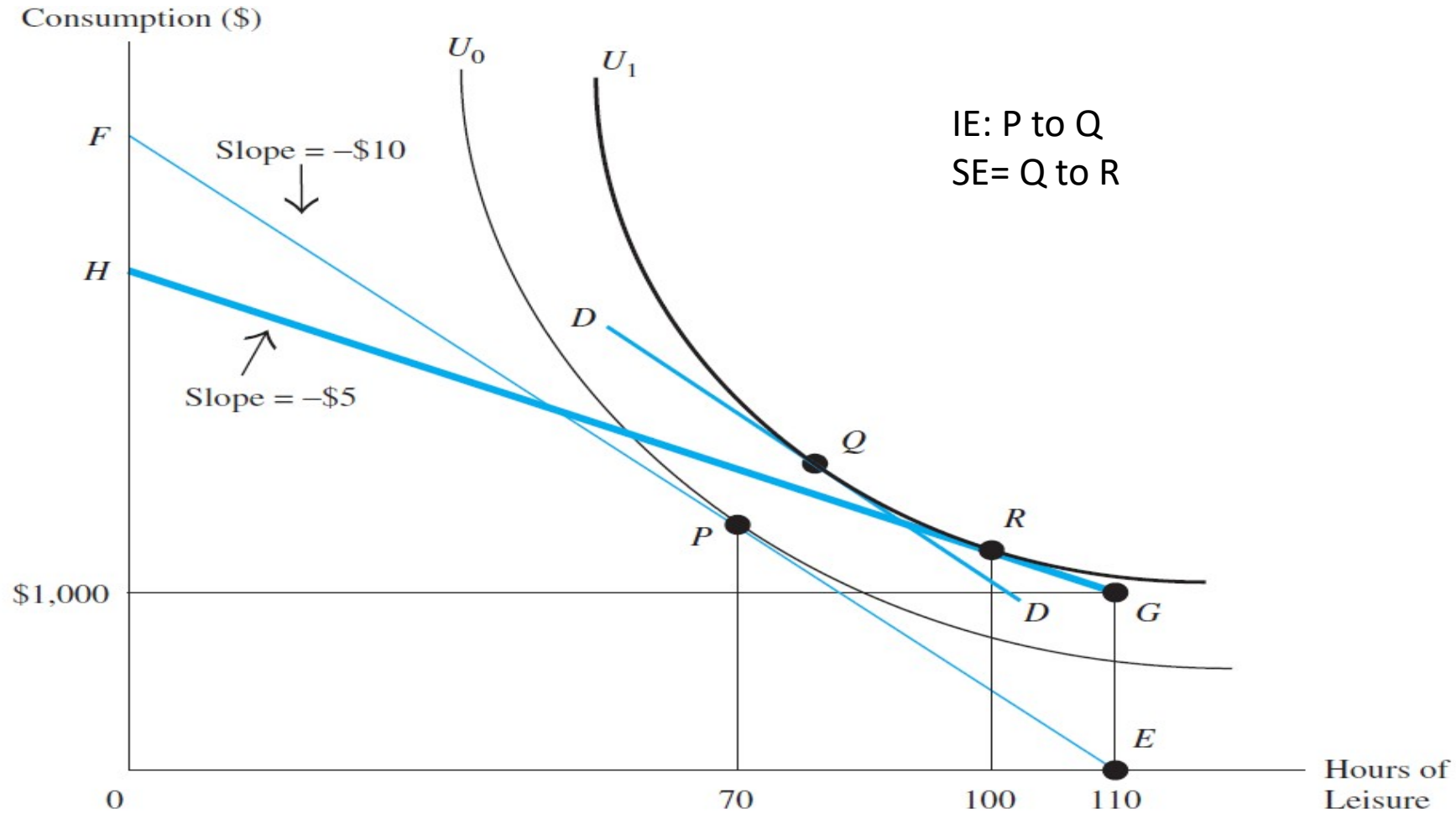
2. The Benefit Reduction Rate

- **One solution** is, instead of take-it-or-leave-it cash grant, use the **benefit reduction rate approach** → reduce the grant amount as the recipients income increases.
- $S = B - tY$, where
 - S - is the amount of grant
 - B - is the maximum amount of support,
 - t - is benefit reduction rate and
 - Y - is income.
- As the person's earned income increases, the amount of subsidy declines, and becomes zero as income keeps on increasing → The **Break-Even Level of Income**.
- The **break-even income** is the level of earned income at which the actual subsidy payment received by an individual or family becomes zero, and dropped from an income maintenance program. $S = B - tY = 0 \rightarrow B = tY$

2. The Benefit Reduction Rate

- Q#: Assuming $B = 1000$, $W = 10$ and $t = 50\%$, show
 1. The effect of the program on L_s relative to
 - A. take-it-or-leave-it cash grant
 - B. Without the support (grant)
 2. Disentangle the IE and SE and show that they reinforce each other to increase leisure time.

2. The Benefit Reduction Rate



2. The Benefit Reduction Rate

- Even this liberal “workfare” program, therefore, seems to have work disincentives because she works fewer hours than she would have worked in the absence of welfare.
- It is difficult to formulate a relatively generous welfare program without substantially reducing work incentives.
- These programs **reduce both**
 1. the probability of a person working and
 2. the number of hours worked by those who remain on the job.

3. The Earned Income Tax Credit (EITC)

- Aims to give public support by with no (or little) impact on working hours.
- Designed to incentivize work.
- Applied in USA
- **Eligibility:**
 1. Work
 2. Earned income is below a certain amount
- **Support:** eligible persons can ask a tax return
- The benefit is a certain percentage of earned income.

3. The Earned Income Tax Credit (EITC)

- It applies d/t approaches: E.g., in USA
 1. First, support (S) = $aY, 0 < a \leq 1$, for $Y \leq Y_1$
 $\rightarrow \text{Total income} = Y + aY$
 2. $S = aY_1$ for $Y_1 < Y < Y_2$
 $\rightarrow \text{Total income} = Y + aY_1$
 3. Then, $S = aY_1 - bY$, for $Y_2 < Y < Y^{max}$ & $aY_1 \leq bY$
 $\rightarrow \text{Total income} = Y(1 - b) + aY_1$
 4. $S = 0$ if $Y > Y^{max}$
 $\rightarrow \text{Total income} = Y$

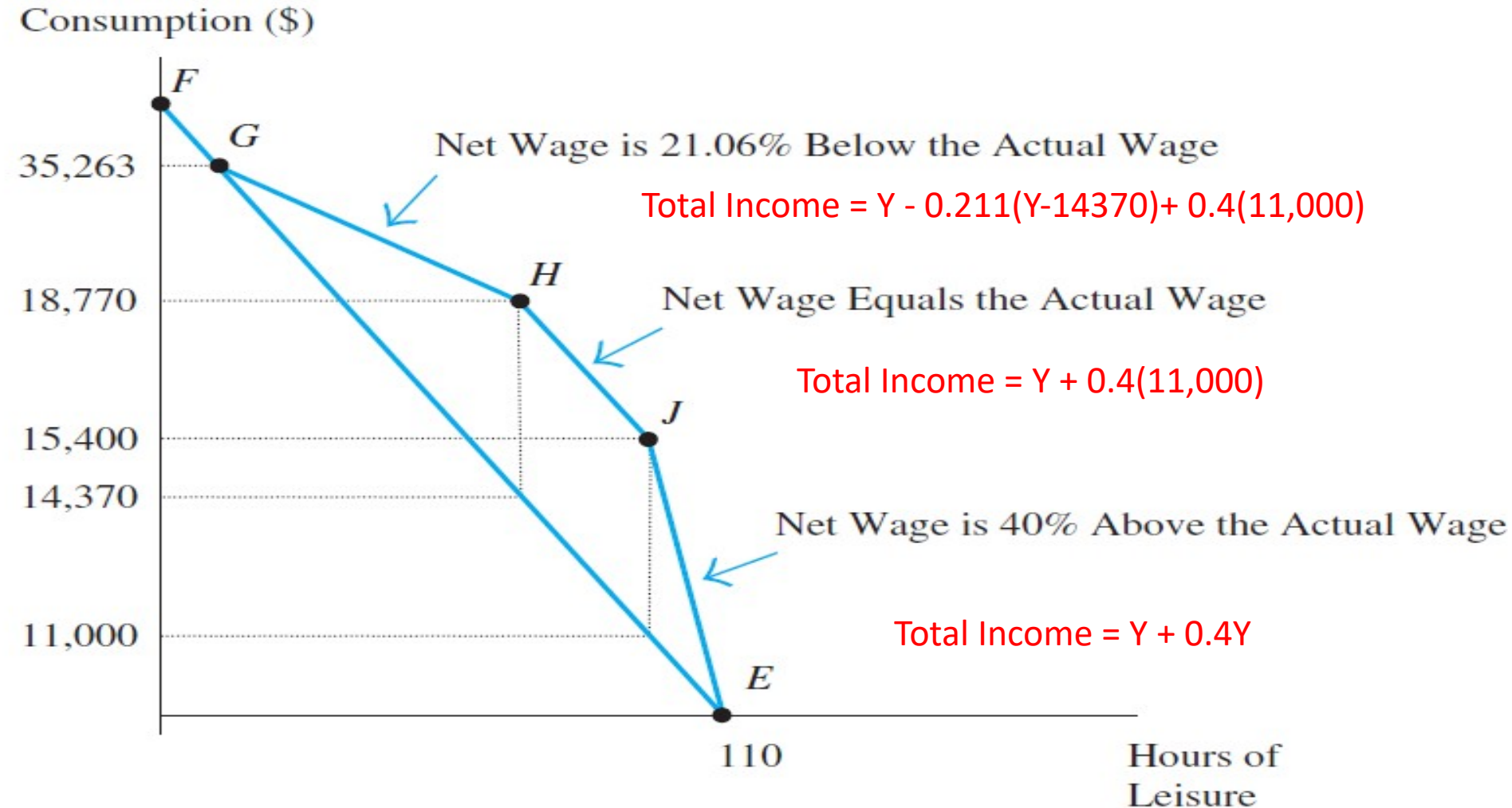
Where Y is wage income, Y_1 , Y_2 and Y^{max} are ceiling values of income determined by the government to provide the support by both reducing cost of the programs and incentivizing work.

- Gives a budget line with many spikes (kinks).

3. The Earned Income Tax Credit (EITC)

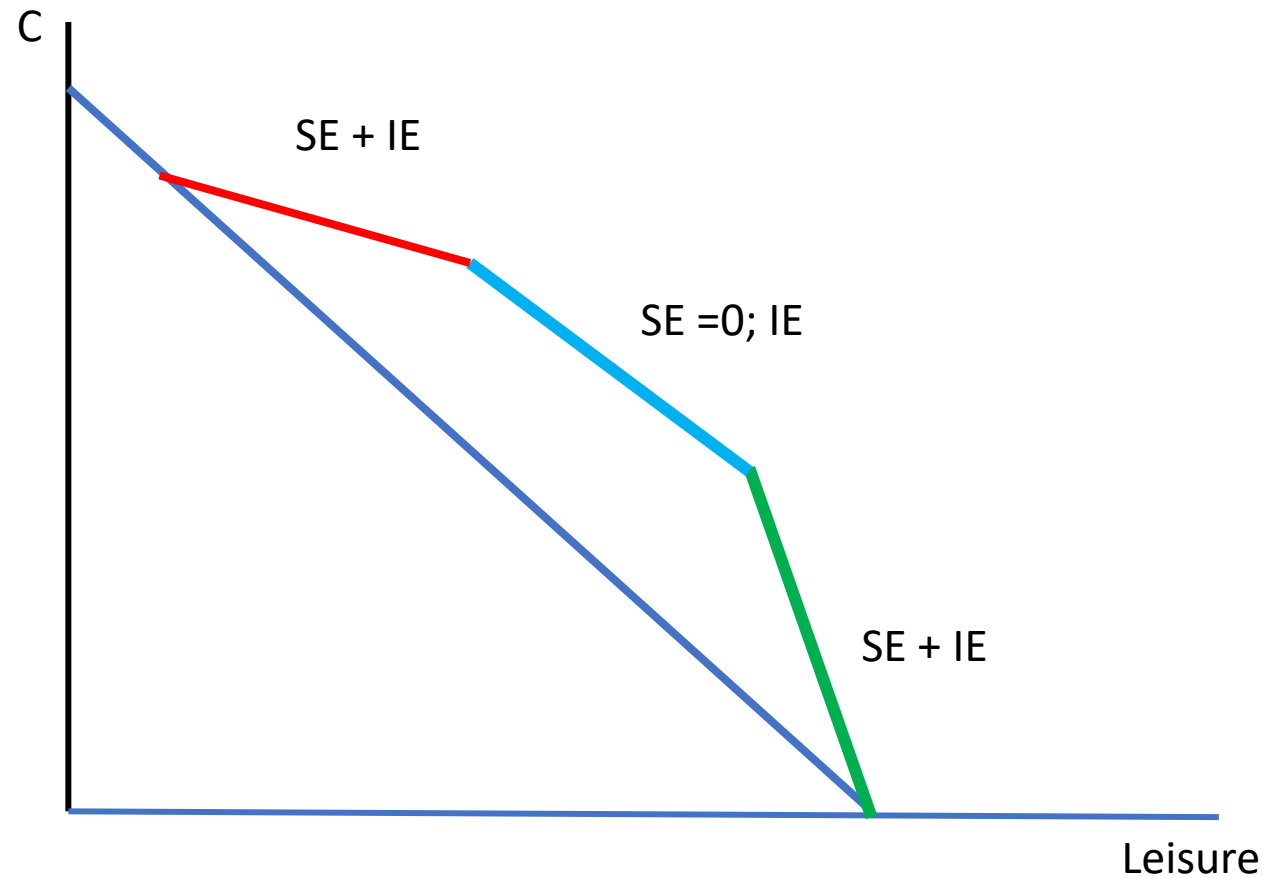
- **Example:** practical examples from USA (Borjas, 2016).
- Eligible person could claim a tax credit of up to 40% ($a = 0.4$) of her earnings as long as she earned less than \$11,000 per year ($Y_1 = 11,000$).
- This maximum credit would be available as long as she earned between \$11,000 and \$14,370 ($Y_2 = 14,370$).
- After reaching the \$14,370 threshold, the credit would begin to be phased out.
- Each additional earned dollar greater than 14,370 reduces the credit by 21.06 cents ($b = 0.2106 \approx 0.211$)
- This formula implies that the credit completely disappears once the woman earns \$35,263 (Y^{max}).
- **Question.**
 1. Draw the budget line and
 2. Discuss the impact of the EITC implications on labor force participation and hours of work.

3. The Earned Income Tax Credit (EITC)



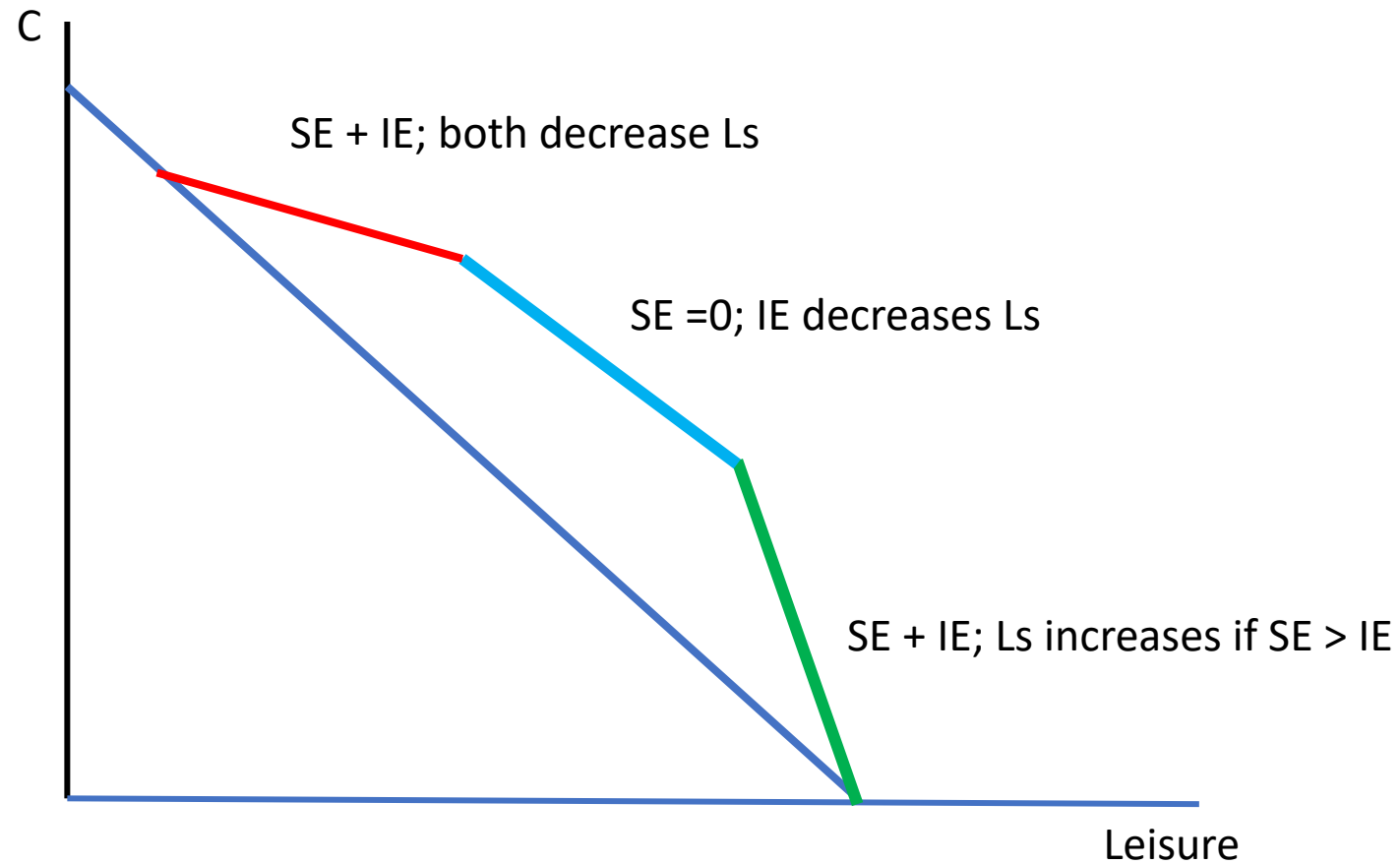
3. The Earned Income Tax Credit (EITC)

- Impact of the program on L_s :

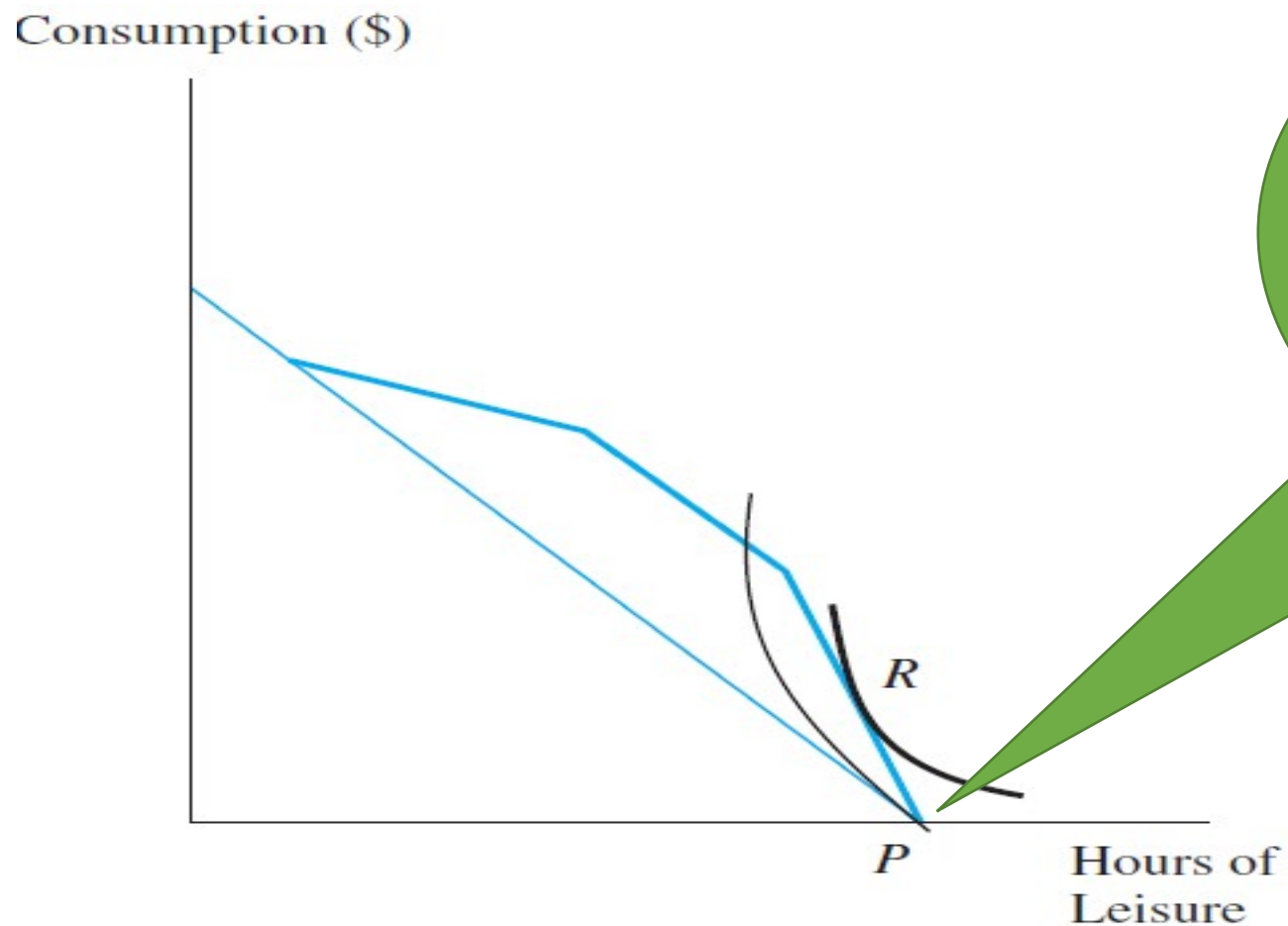


3. The Earned Income Tax Credit (EITC)

- Impact of the program on L_s :



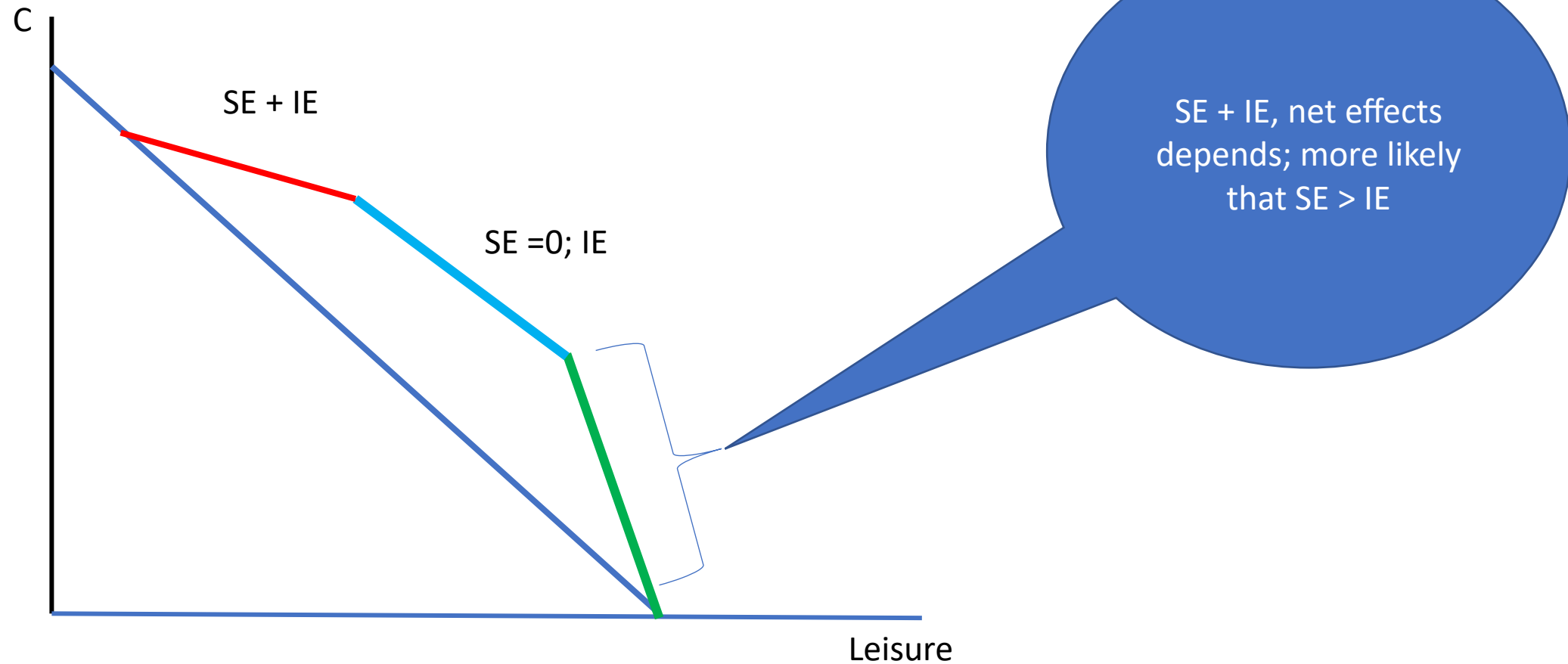
EITC impact on hours of work



Not working is inferior when
there is EITC.
So, EITC increases labor force
participation

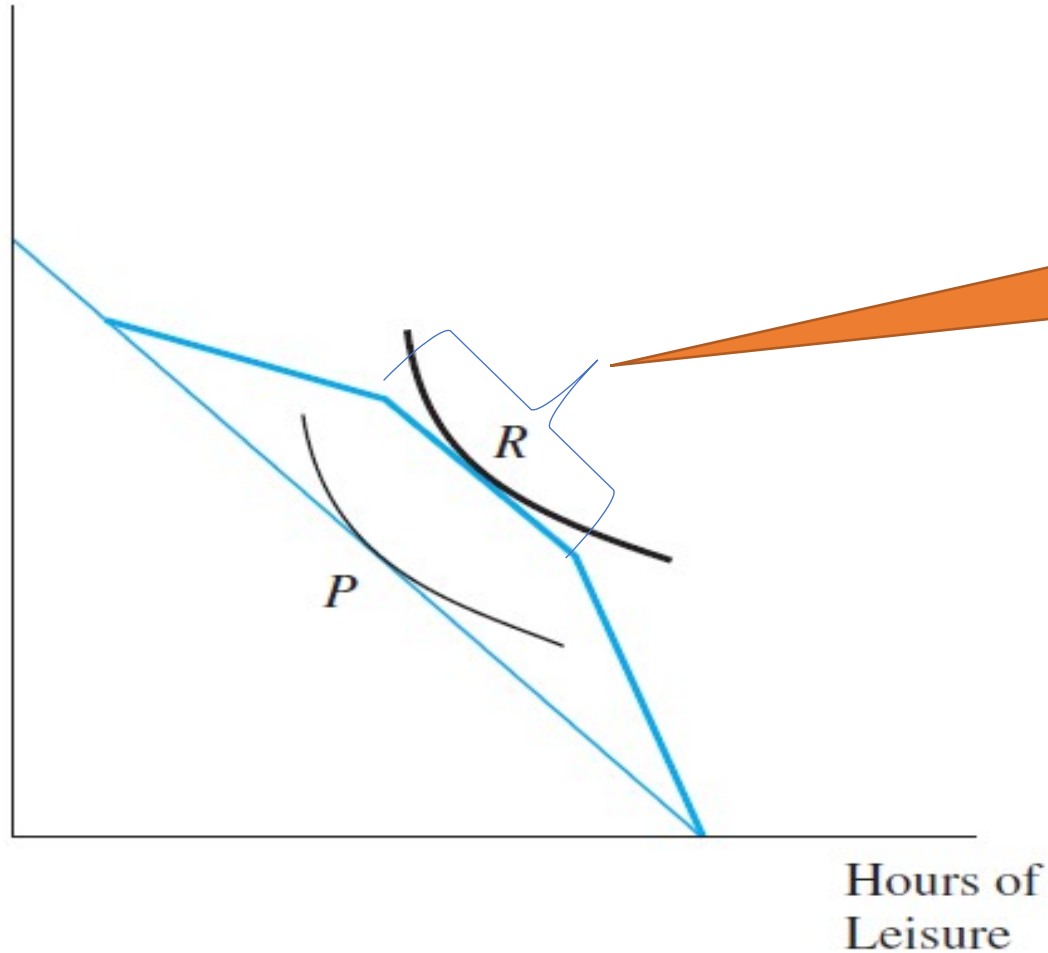
3. The Earned Income Tax Credit (EITC)

- Impact of the program on L_s :



EITC impact on hours of work

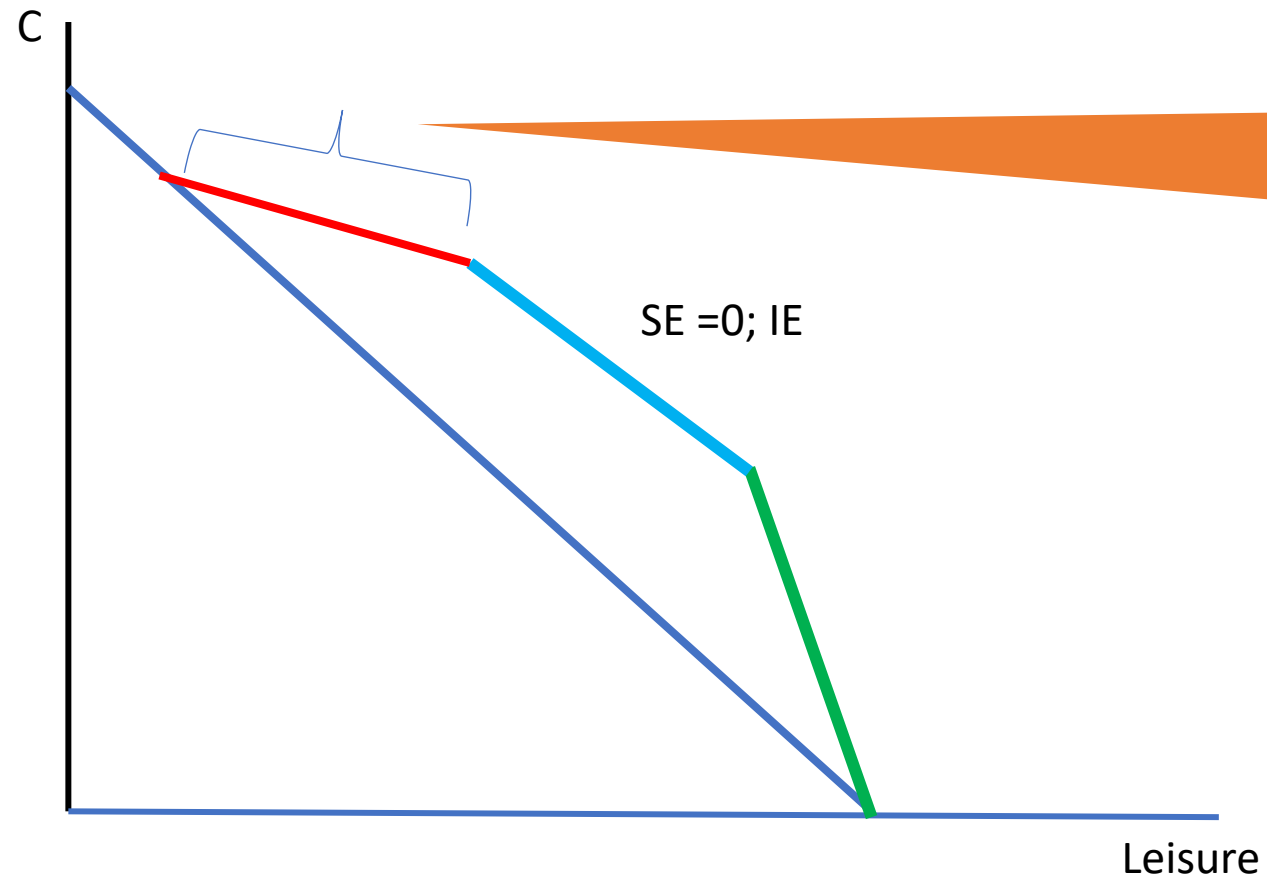
Consumption (\$)



EITC has only income effect → EITC reduces hours of work

3. The Earned Income Tax Credit (EITC)

- Impact of the program on L_s :



EITC impact on hours of work

- Prediction of the Theory:

1. EITC increases labor force participation rate of low-wage workers since the tax credit is granted only to persons who work, more persons will enter the labor force to take advantage of this program.
2. EITC may change the number of hours worked by persons who would have been in the labor force even in the absence of the program. It may reduce hours of work for at least some group of the workers. The net impact depends on which of the SE and IE dominates.

Empirical evidence:

- The available evidence confirms the theoretical prediction that the EITC draws many new persons into the labor force.

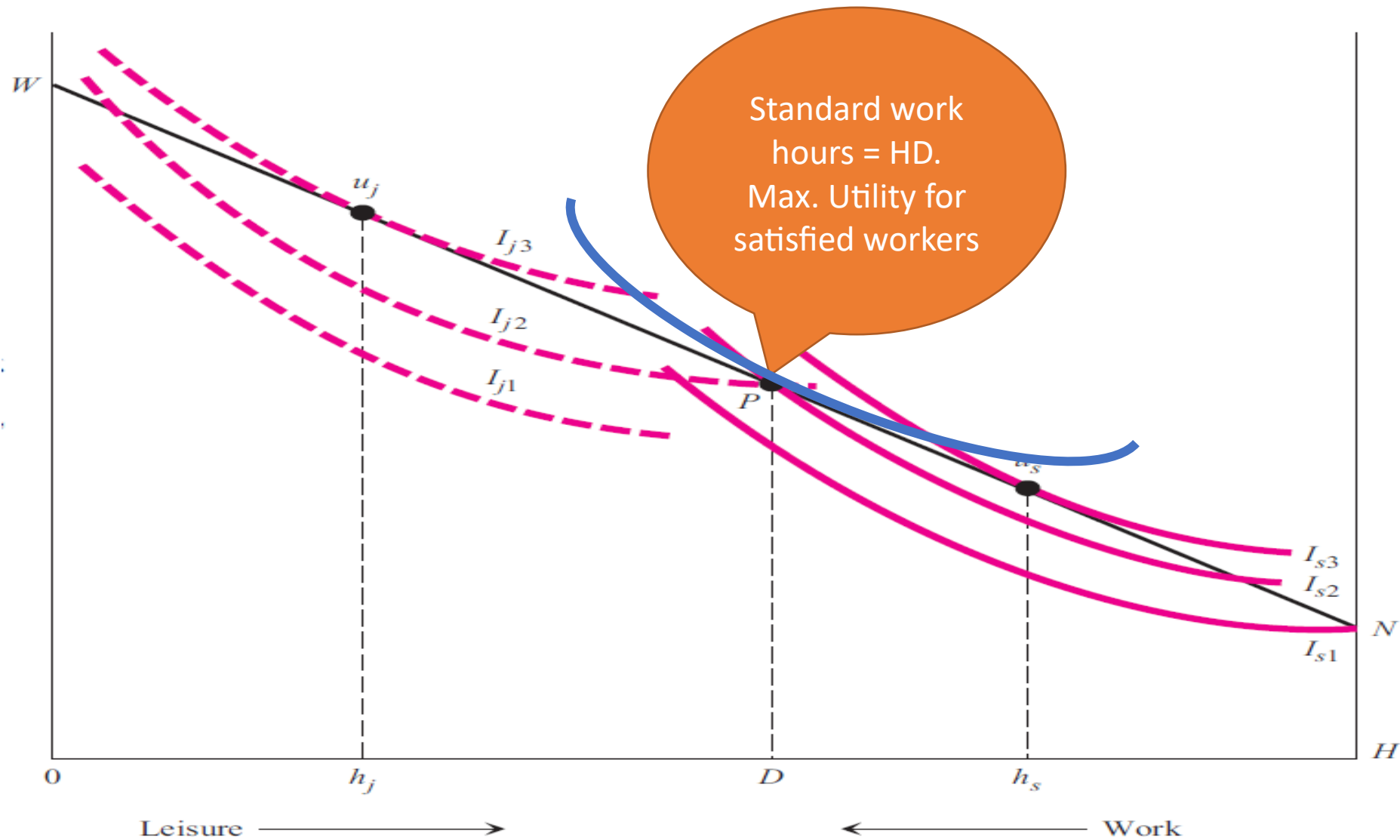
Other mechanisms to reduce undesired impact of public support programs

1. Limit the period (time) of support so that individuals who are able to work will likely work, reserving the support for future unforeseen income shocks while the needy ones will demand the support anyway.
2. Reduce the benefit reduction rate, but it increases the cost of the program and makes more people eligible for the support.
3. Demand beneficiaries to provide public work so that persons who are able to work may prefer employment if the paid work pays more.
4. Work requirements to get support – after providing support for a given period, some gov't/companies induce recipients to work some hours per week to get the support. If the beneficiary couldn't get a paying job,
 - s/he can work for free, which may induce the employer to hire her/him if they are pleased with the performance.
 - Attain trainings that increase skill and lead to employment
 - Attend school

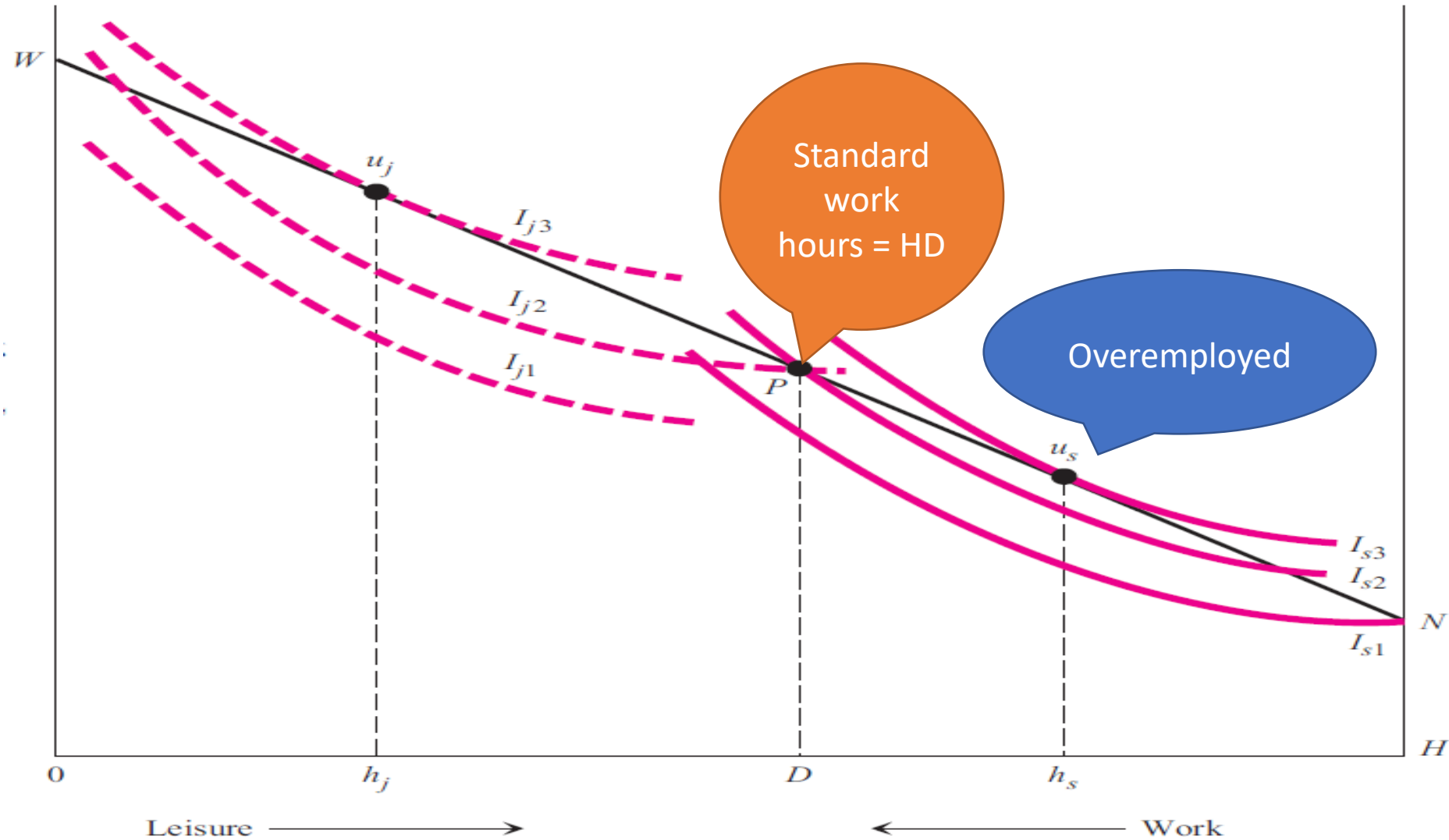
Standard Working Hours Requirement

- Our discussion so far has implicitly assumed that workers can **individually determine the number of hours they work**. **This is typically not the case**.
- E.g. working hours in Ethiopia 40 - 48 hours per week. Employers need to pay for hours worked > 48 hours per week. This is the same in many countries.
- This standard (fixed) hours of work may result in **#3 groups of workers**
 1. **Overemployed:**
 - A. for individuals who prefer to work less hours than the standard hours
 - B. May lead to **absenteeism**; he may more or less habitually miss a day of work every week or so. For instance, the ratio of full-time workers with absences in a typical week to total full-time employment—was 3.2 percent in 2006.
 - C. Labor turnover
 2. **Underemployed:**
 1. Prefer to work more hours
 2. May lead to moonlighting – taking a second job. In fact, in 2005 approximately 5.3% of all employees—held multiple jobs.
 3. **Satisfied with the standard work** – most of the worker.

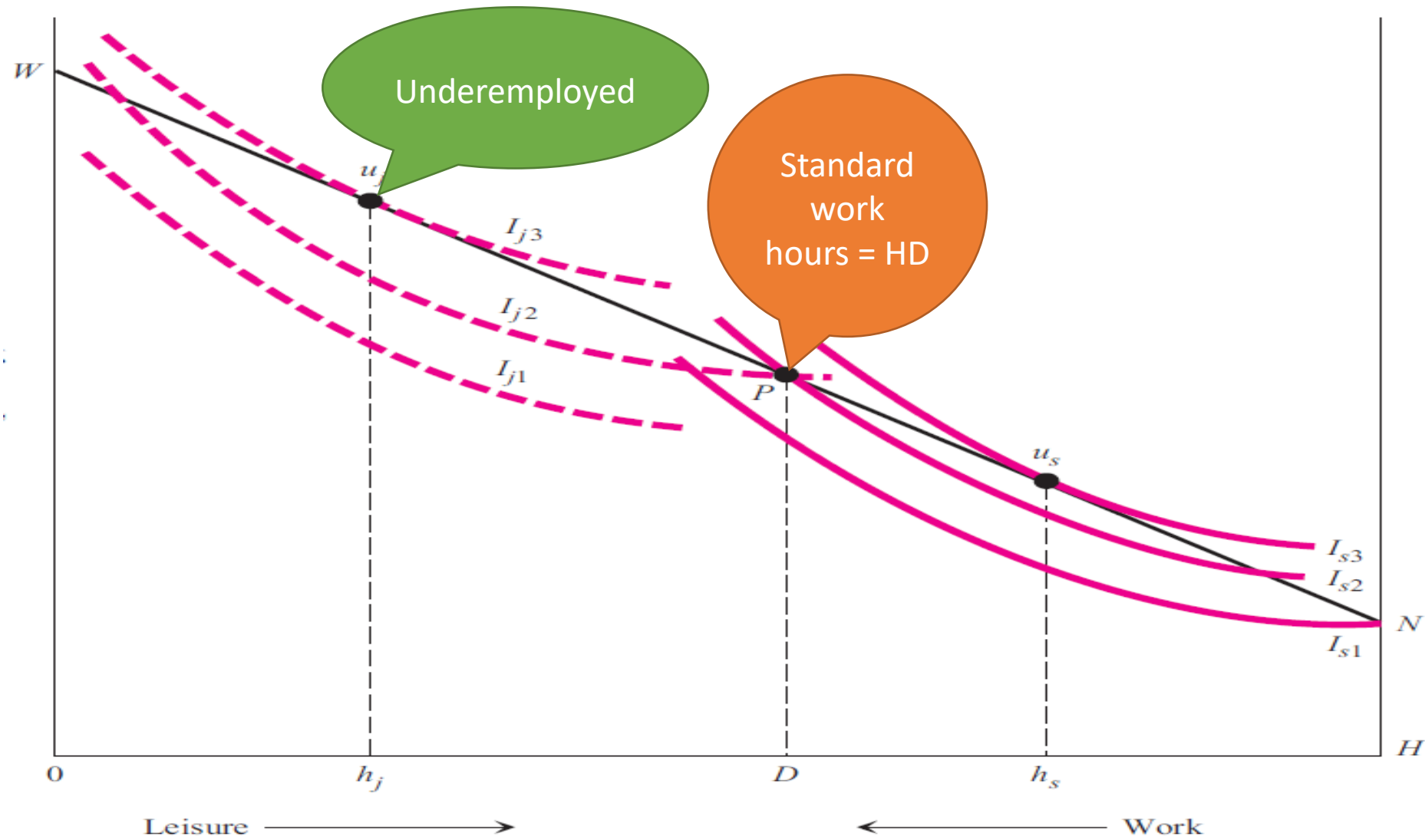
Standard Working Hours Requirement



Standard Working Hours Requirement



Standard Working Hours Requirement



Premium Wage & Straight Time Equivalent Pay

A. **Premium pay** refers to the higher wages given to employees who work less desirable hours (e.g., night, holidays, evening) or for overtime standard working hours.

Example, **Ethiopia**

- Normal hours of work shall **not exceed** 8 hours a day or **48 hours a week**.
- **In principle, overtime works are prohibited** with exceptions in the occurrence of acute or imminent accident, force majeure, urgent work and substitution of absent workers assigned on work that runs continuously without interruption, in that workers are **required** to work overtime, but at higher wage rate
- Overtime work shall **not exceed four hours** in a day and **12 hours in a week**.
- Premium rate $1.5*W$ (for work b/n 6 am and 10 pm) and $1.75*W$ (for work b/n 10 pm and 6 am), $2*W$ (weekly rest days), $2.5*W$ (public holidays)

B. **Straight Time Equivalent Pay** – identical wage rate paid daily/weekly or monthly.

Straight Time Equivalent Pay

B. Straight Time Equivalent Pay – identical daily or weekly/monthly income from the same number of hours of work.

- **Question:** Does it matter to pay \$6 per hour for the first 8 hours of work and \$9 per hour for an additional 2 hours of overtime work or to pay \$6.60 per hour for each 10 hours of work?

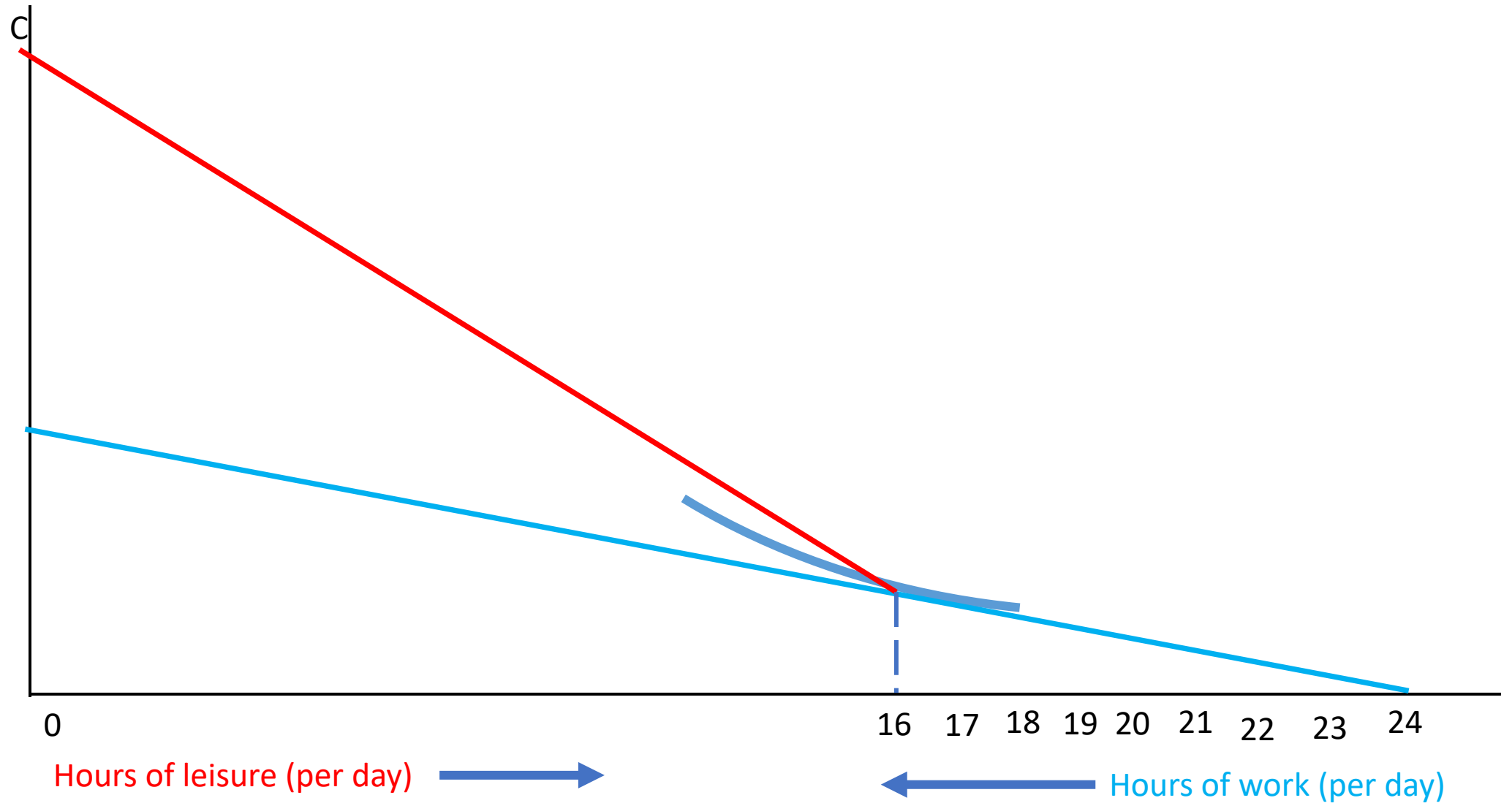
Premium Wage Vs Straight Time Equivalent Pay

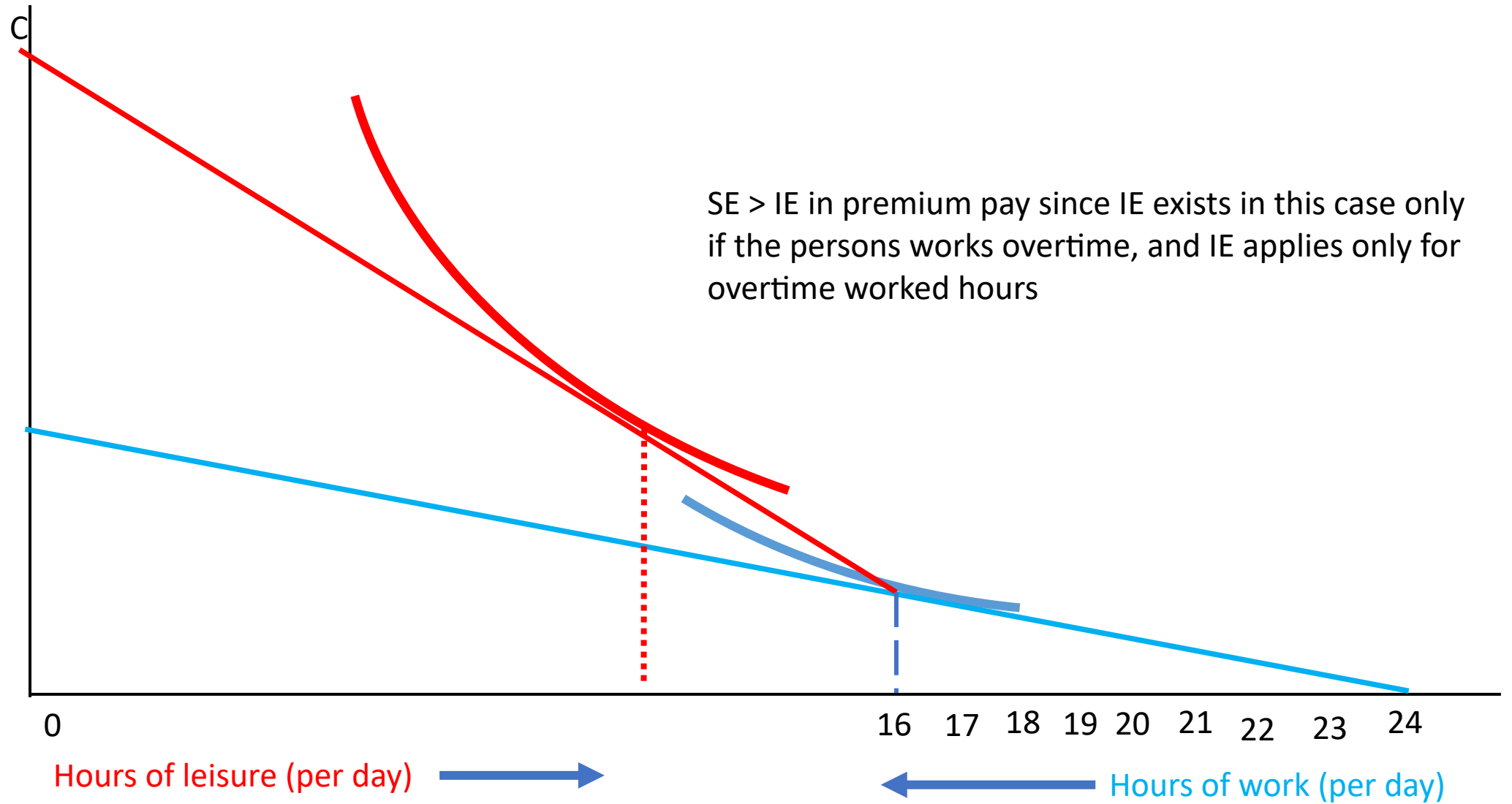
- Both payment plans yield the same daily income of \$66, but payment arrangement differs.
- So one is inclined to conclude that it makes no difference, **but it does make a difference.**
- This is because the two payment arrangements trigger different value for leisure, i.e., the

IE (premium pay) < IE (Straight-time equivalent pay)

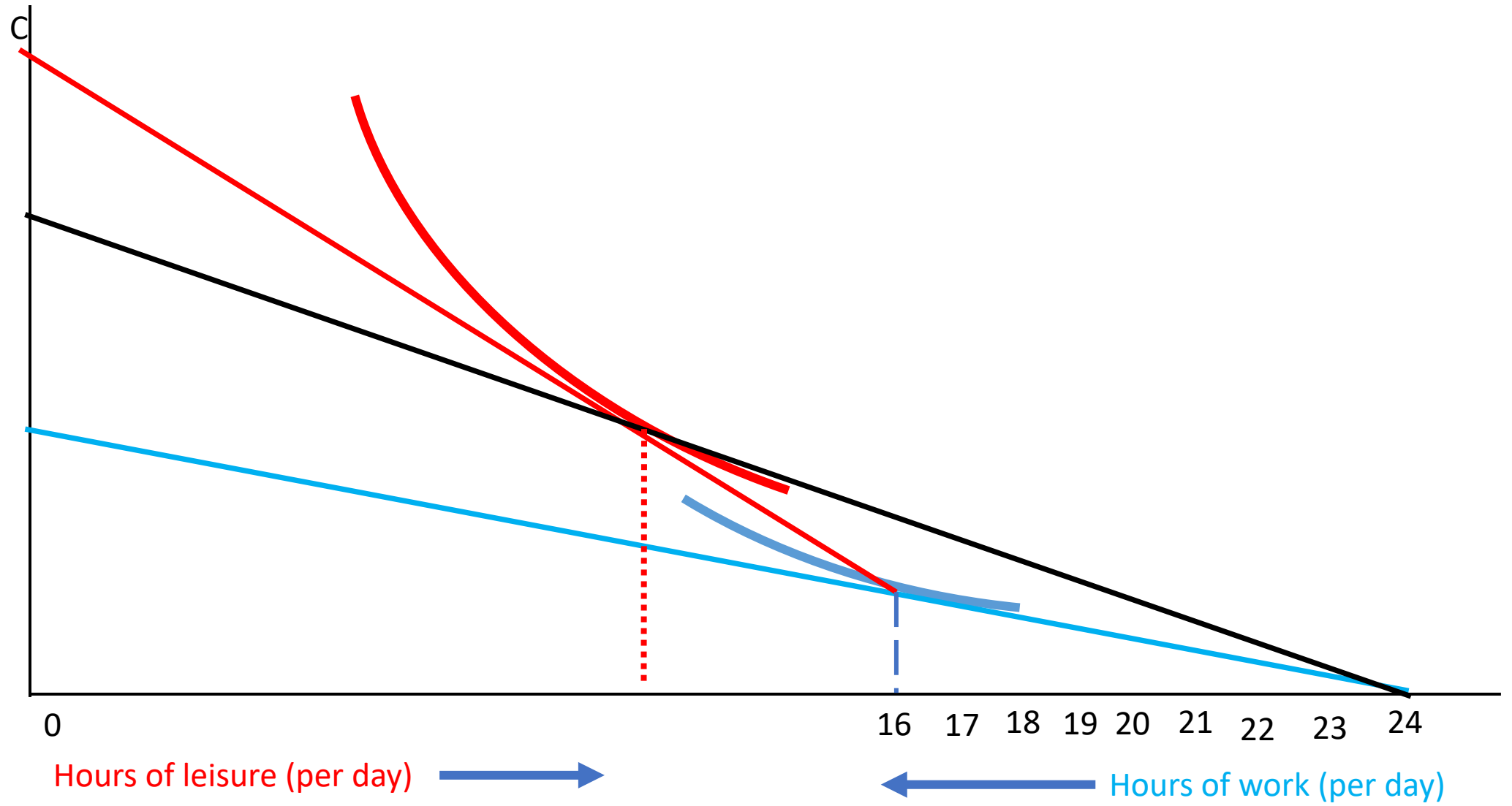
- **Question:**
 - Why?
 - Show graphically?

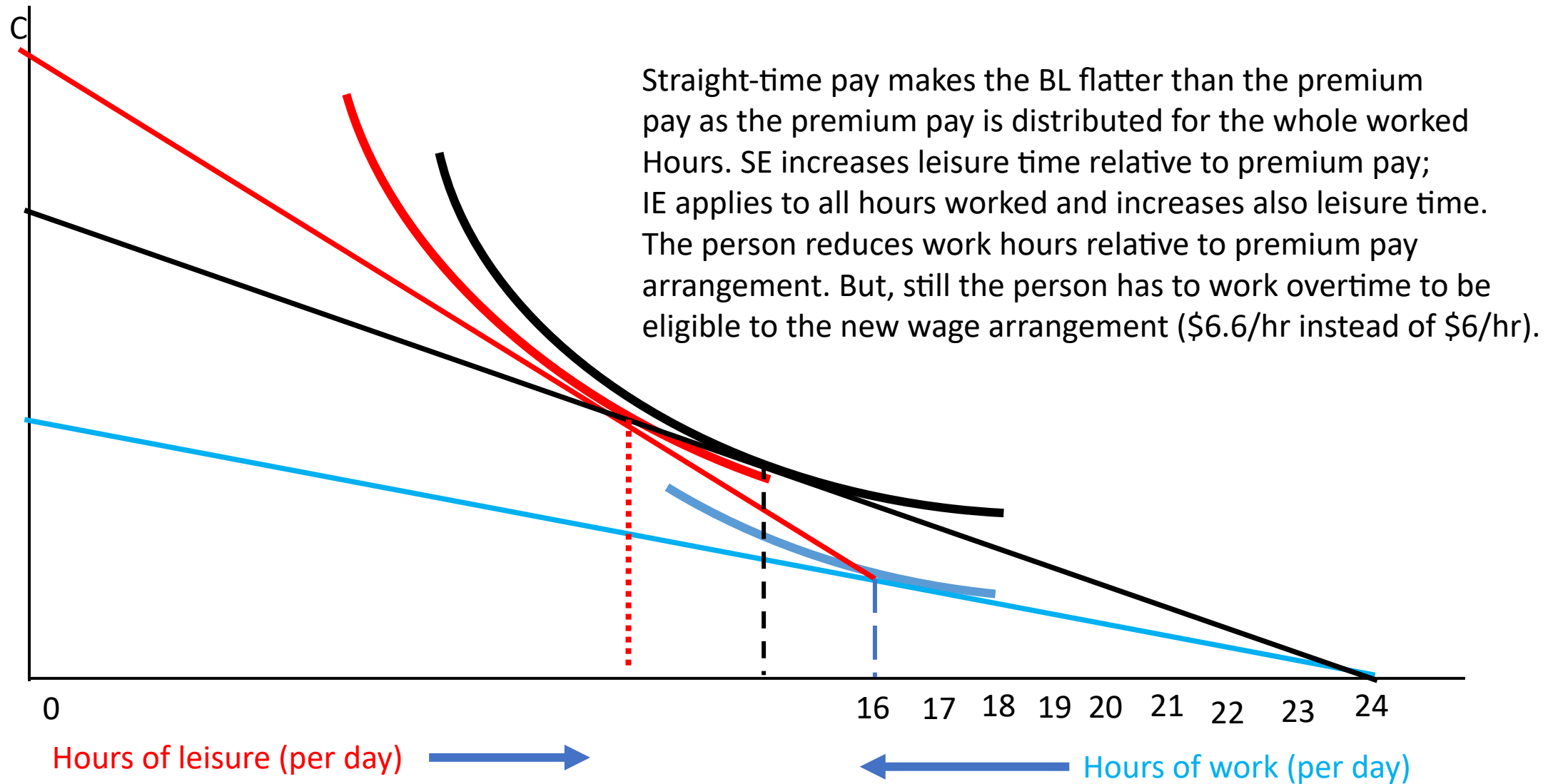
Premium Wage Vs Straight Time Equivalent Pay





Premium Wage Vs Straight Time Equivalent Pay





Premium Wage Vs Straight Time Equivalent Pay

- Premium wage rates for overtime work will call forth more hours of work than a straight-time wage rate that yields the same income at the same number of hours as that actually chosen by an individual paid the overtime premium.
- The use of premium pay will have a relatively small income effect because it applies only to hours worked in excess of standard hours .
- In comparison, the straight-time equivalent wage will have a much larger income effect because it applies to all hours of work.
- This is the labor market **analog of price discrimination** in the product market. Sellers of some products can obtain more revenue by charging different prices for different quantities of output.
- Similarly, an employer can obtain a greater amount of labor for a given outlay by paying different wage rates for different hours of work.

2.5. Labor Mobility

1. Determinants of Labor Mobility
2. Labor Market Effects of Migration

2.5. Labor Mobility

Labor Mobility –

- Movement of workers from one employer to another
- The movement can be between:
 1. Employers
 2. Locations
 3. Occupation
 4. Any combination of the above three
- It is one of the striking features of labor markets.

2.5. Labor Mobility

- **Determinants of mobility**

1. Age – ceteris paribus, the older that a person is, the less likely he or she is to migrate.
2. Family Factors - The potential costs of migrating multiply as family size increases. Ceteris paribus, married workers have less tendency to migrate than single people.
3. Education - The higher one's educational attainment, all else being equal, the more likely it is that one will migrate.
4. Distance - The probability of migrating varies inversely with the distance a person must move.
5. Unemployment Rates – (1) Families headed by unemployed people are more likely to migrate than others , and (2) the rate of unemployment at the origin positively affects out-migration
6. Other Factors –
 - i. Language spoken at the destination
 - ii. home ownership deters migration
 - iii. State and local government policies may influence labor migration (e.g., tax rate)
 - iv. location characteristics

Labor Market Effects of Migration

- Labor mobility contributes to **allocative efficiency** by relocating labor resources away from lower-valued and toward higher-valued employment.
- Under certain conditions, workers of a given type will relocate until the **net value of the marginal product of labor** (VMP) is the same in all similar employments.
- Immigrants may have an adverse impact on the job opportunities of the native workers whose skills resemble those of the immigrants
- Immigration of high-skill workers can be hugely beneficial.
- High-skill immigrants generate human capital externalities. The presence of high-skill immigrants exposes natives to new forms of knowledge, increases their human capital, and makes them more productive.